

CHY1009

Chemistry and Environmental Studies

Module 2

Part 3

States of Matter, Thermodynamics, Kinetics and Catalysis

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Thermodynamics

- *Flow of heat*
- The branch of physical chemistry which deals with the energy changes accompanying a chemical reaction

It establishes an *relationships between* *heat* and *work*.

System and Surrounding

- For the purposes of physical chemistry, the universe is divided into two parts, the system and its surroundings.
- The system is the part of the world in which we have a special interest. It may be a reaction vessel, an engine, an electrochemical cell, a biological cell, and so on.
- The surroundings comprise the region outside the system and are where we make our measurements.

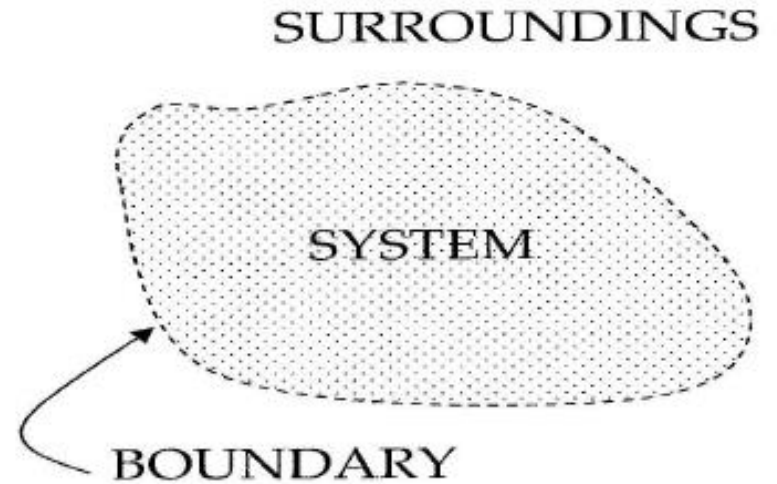
Thermodynamics

- **Types of system:**

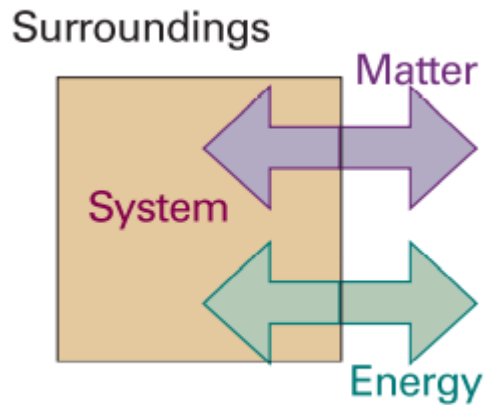
Open system

Closed system

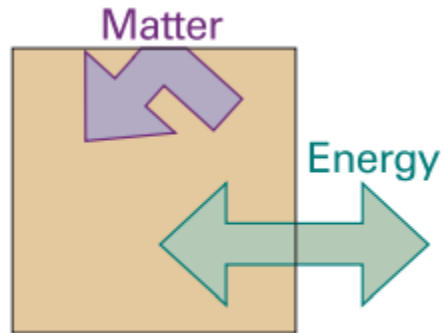
Isolated system



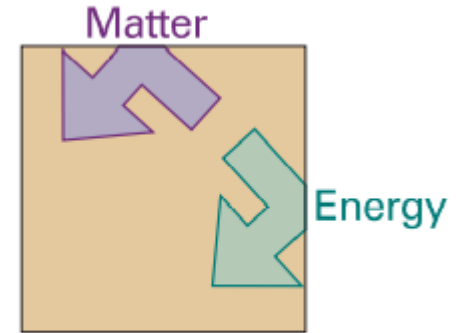
Different types of systems



(a) Open



(b) Closed



(c) Isolated

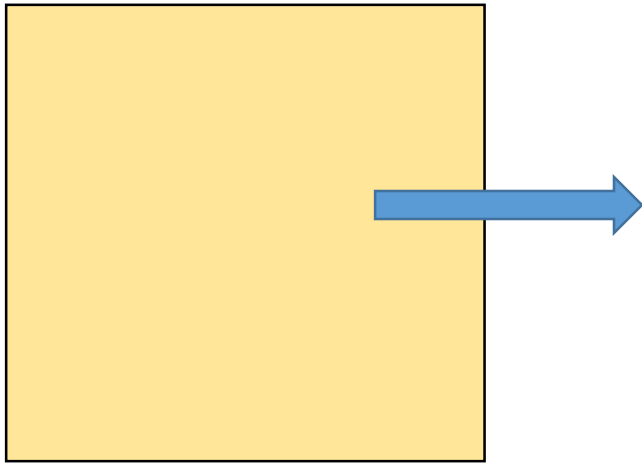
An open system can exchange **matter and energy** with its surroundings.

A closed system can exchange **energy** with its surroundings, but it cannot exchange **matter**.

An isolated system can exchange **neither energy nor matter** with its surroundings.

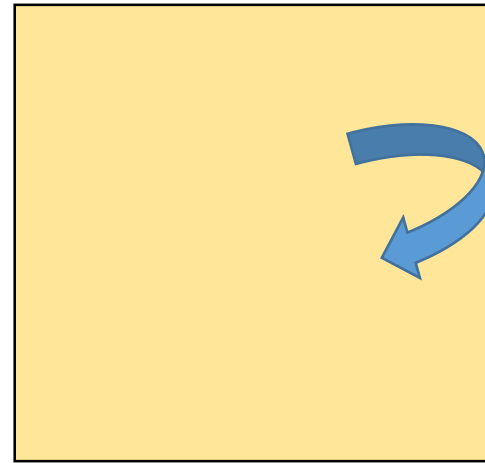
Diathermic & Adiabatic Wall

Diathermic wall



Passage of heat is
permitted

Adiabatic wall



Passage of heat is
not permitted

Types of Thermodynamic Processes

Isobaric



Boiling of a water in **open** vessel

Isochoric



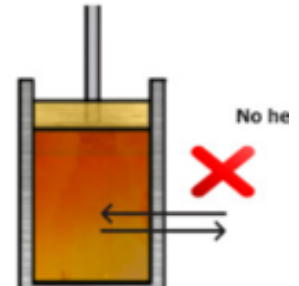
Boiling of a water in **closed** vessel

Isothermal



Boiling of a water at constant Temperature

Adiabatic



Physical properties of a system

- **Thermodynamic process**

Isothermal process : Temperature of the system remains constant: $\Delta T = 0$

Adiabatic process : No exchange of heat

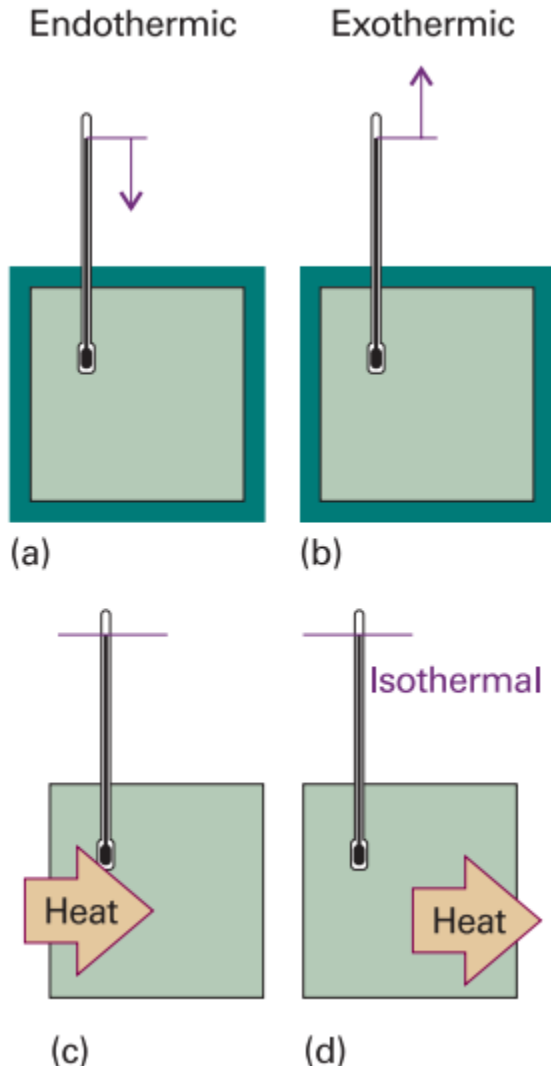
Isobaric process: Pressure of the system remains constant: $\Delta P = 0$

Isochoric process : Volume of the system remains constant: $\Delta V = 0$

Reversible process :

Irreversible process :

Exothermic and endothermic process



Endothermic – Gaining of heat during a process

Exothermic – Release of heat during a process

Adiabatic system:

Heat neither goes out or come in.

- Temperature reading in thermometer falls during endothermic process.
- Temperature rises during exothermic process.

Isothermal Process; Heat is exchanged between system and surrounding depending upon the process as may be.

So, T of the system is maintained constant.

Physical properties of a system

Intensive Properties

- does not depend on the system size or the amount of material in the system

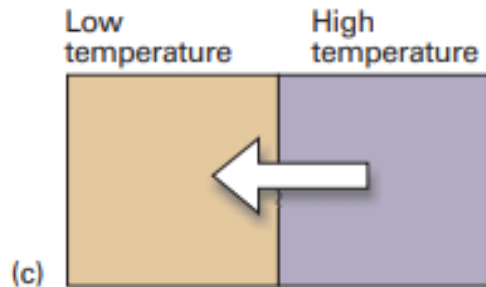
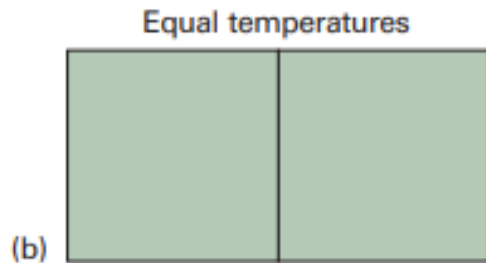
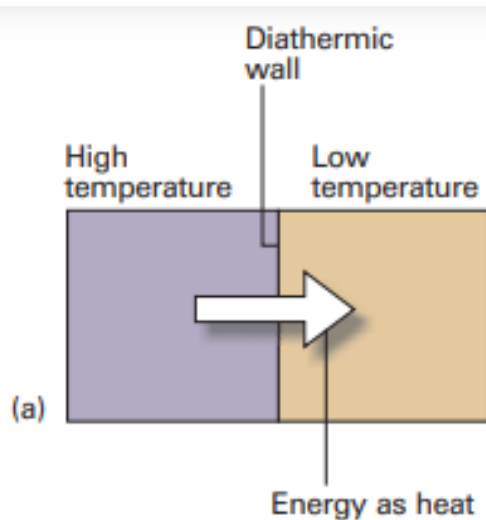
Temperature,
Pressure, viscosity,
surface tension,
refractive index,
specific heat,
Density

Extensive Properties

- Depend on the system size or the amount of material in the system

mass,
volume
energy
heat capacity,
entropy, Gibb's free energy

Objects in thermal equilibrium



The temperature, T , is the property that indicates the direction of the flow of energy through a thermally conducting rigid wall.

If energy flows from **A to B** when they are in contact, then we say that **A has a higher temperature than B**

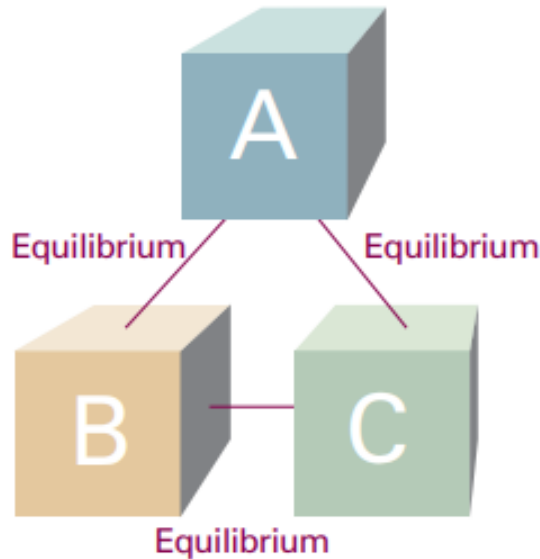
Vice versa, If energy flows from **B to A** when they are in contact, then we say that B has a higher temperature than A.

Thermal equilibrium is established if no change of state occurs when two objects A to B are in contact through a diathermic boundary.

The Law's of Thermodynamics

- Zeroth Law
- First Law
- Second Law
- Third Law

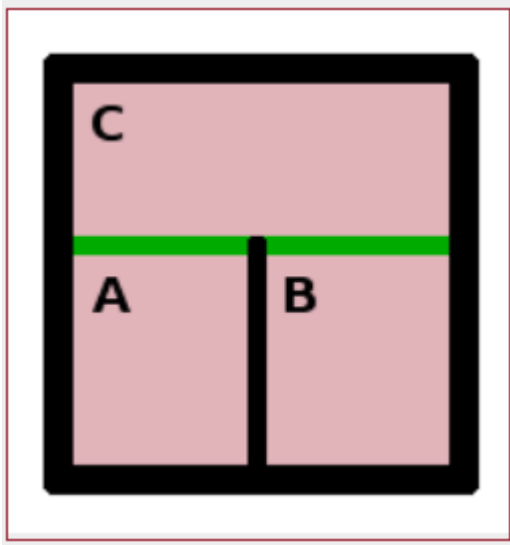
Zeroth Law of Thermodynamics



- Suppose an object **A** (which we can think of as a block of iron) is in **thermal equilibrium with an object B** (a block of copper), and that B is also in thermal equilibrium with another object **C** (a flask of water). Then it has been found experimentally that A and C will also be in thermal equilibrium when they are put in contact.
- This observation is summarized by the Zeroth Law of thermodynamics:

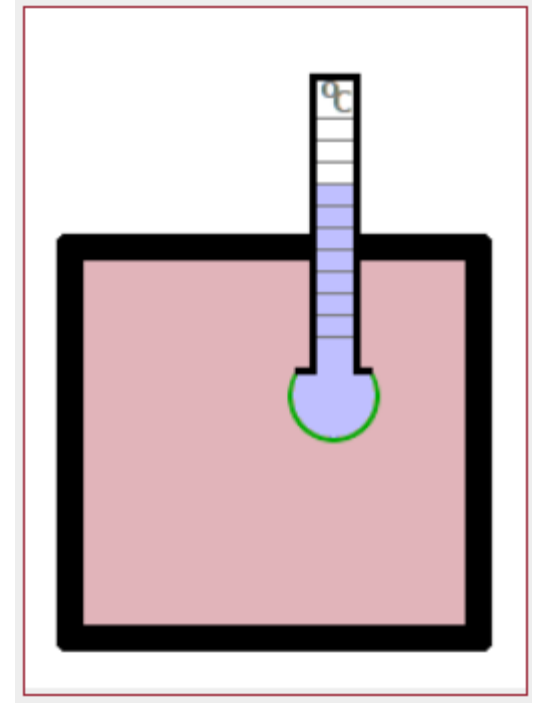
“If A is in thermal equilibrium with B, and B is in thermal equilibrium with C, then C is also in thermal equilibrium with A”

Zeroth Law of Thermodynamics



— Adiabatic wall

— Diathermal wall



If $T_A = T_C$ and $T_B = T_C$

Then, $T_A = T_B$

Zeroth Law Provides the Basis for the thermometer function.

Where mercury and the liquid in contact are separated by a diathermal wall

Application of Zeroth Law of Thermodynamics



Measuring the atmospheric temperature



Temperature measurement in Air Conditioner

What did you find from the zeroth law of thermodynamics?



First Law of Thermodynamics

- *It is law of conservation of energy*
- ***Energy cannot be created or destroyed***, it can only be transformed from one form to another form.
- The concept of internal energy (E or U), Heat (q), Work(w), and enthalpy (H) follows from first law of thermodynamics.

Work

- The fundamental physical property in thermodynamics is work
- *Work is motion against an opposing force*. Doing work is equivalent to raising a weight somewhere in the surroundings.
- An example of doing work is the expansion of a gas that pushes out a piston and raises a weight (*Expansion work or Pressure-Volume Work*). –Pressure-volume change occurs
- A chemical reaction that drives an electric current through a resistance also does work, because the same current could be driven through a motor and used to raise a weight (*Non expansion work*). –Gibbs free energy change occur

Energy

- *The energy of a system is its capacity to do work.*
- When work is done on an otherwise isolated system (for instance, by compressing a gas or winding a spring), the capacity of the system to do work is increased; in other words, the energy of the system is increased.
- When the system does work (when the piston moves out or the spring unwinds), the energy of the system is reduced and it can do less work than before.

Heat

- Experiments have shown that the *energy of a system may be changed by means other than work* itself.
- When the *energy of a system changes as a result of a temperature difference between the system and its surroundings* we say that energy has been transferred as heat.

Molecular/atomic motion during energy transfer to or work done on surrounding

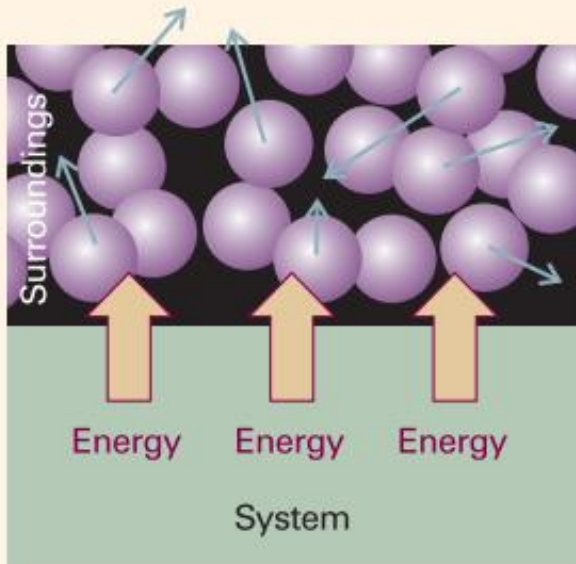


Fig. 2.3 When energy is transferred to the surroundings as heat, the transfer stimulates random motion of the atoms in the surroundings. Transfer of energy from the surroundings to the system makes use of random motion (thermal motion) in the surroundings.

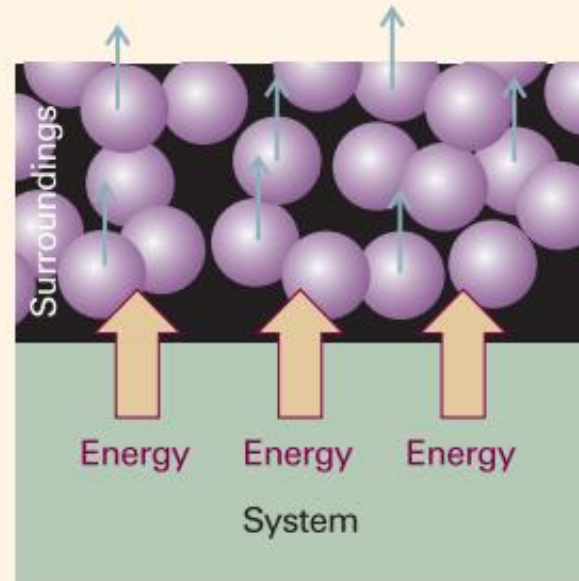


Fig. 2.4 When a system does work, it stimulates orderly motion in the surroundings. For instance, the atoms shown here may be part of a weight that is being raised. The ordered motion of the atoms in a falling weight does work on the system.

State Functions

A **state function** is a property whose value does not depend on the path taken to reach that specific value.

From first law of thermodynamics

Internal Energy (E),
Enthalpy (H),

From second law of thermodynamics

Entropy (S),
Free Energy (G)

Internal Energy (E)

Internal energy

- Every substance possesses a definite amount of energy which depends upon its chemical nature, temperature, pressure and volume.

$$E_{\text{total}} = E_t + E_r + E_v + E_e + E_n + E_i$$

$$\Delta E = E_2 - E_1$$

Internal energy

- The total energy of a system is called its internal energy, **U**.
- The internal energy is the total kinetic and potential energy of the molecules in the system.

$$\Delta U = U_f - U_i$$

Where, ΔU - change in internal energy ;

U_i - internal energy of initial state;

U_f internal energy of final state

Changing state variable such as Pressure (**P**), Temperature(**T**), Volume (**V**) will alter the internal energy of a system.

$$w = - p dV$$

(doing work on the system changes the internal energy of a system)

Internal energy

- It has been found experimentally that the internal energy of a system may be changed either by doing work on the system or by heating it.

$$\Delta U = q + w$$

$$w = - p \, dV$$

q = heat change

w = work

If the system absorbs heat $q > 0$

If system loses heat $q < 0$

If system does work on surrounding $w < 0$

If work is done on the system $w > 0$

Internal energy

$$\Delta U = q + w$$

q	+	w	=	ΔU
---	---	---	---	------------

+		+		+
---	--	---	--	---

+		–		Depends on <i>sizes</i> of q and w
---	--	---	--	--

–		+		Depends on <i>sizes</i> of q and w
---	--	---	--	--

–		–		–
---	--	---	--	---

*For q : + means system *gains* heat; – means system *loses* heat.

For w : + means work done *on* system; – means work done *by* system.

The sign convention of q and w

Illustration 2.1 *The sign convention in thermodynamics*

If an electric motor produced 15 kJ of energy each second as mechanical work and lost 2 kJ as heat to the surroundings, then the change in the internal energy of the motor each second is

$$\Delta U = -2 \text{ kJ} - 15 \text{ kJ} = -17 \text{ kJ}$$

Suppose that, when a spring was wound, 100 J of work was done on it but 15 J escaped to the surroundings as heat. The change in internal energy of the spring is

$$\Delta U = +100 \text{ kJ} - 15 \text{ kJ} = +85 \text{ kJ}$$

Units of Internal Energy, Heat, and Work

- U, q, and w are all measured in the same units, the joule (**J**).
- The joule, named after the nineteenth-century scientist J.P. Joule, is defined as **$1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2}$**
- Changes in **molar internal energy**, ΔU_{m} , are typically expressed in kilojoules per mole (**kJ mol^{-1}**).

Units of Internal Energy, Heat, and Work

- Calories (**cal**) and kilocalories (**kcal**) are still encountered.

$$1 \text{ cal} = 4.184 \text{ J}$$

- An energy of 1 cal is enough to raise the temperature of 1 g of water by 1°C

Expansion work

For infinitesimal change of state

$$dU = q + w$$

Work = Force x displacement

$$dw = -F dz$$

$$dV = A dz$$

$$F = P_{\text{ex}} A$$

$$dw = -P_{\text{ex}} dV \quad w = - \int_{V_i}^{V_f} P_{\text{ex}} dV$$

For expansion w is negative, as $V_f > V_i$

For compression w is positive, as $V_f < V_i$

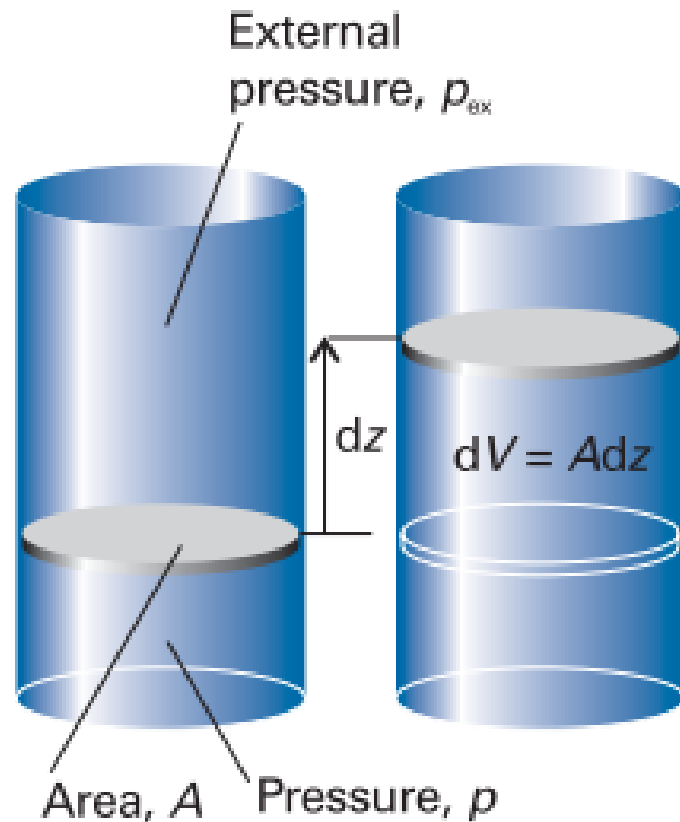


Fig. 2.6 When a piston of area A moves out through a distance dz , it sweeps out a volume $dV = Adz$. The external pressure p_{ex} is equivalent to a weight pressing on the piston, and the force opposing expansion is $F = p_{\text{ex}} A$.

Different types of expansion

(i) Free expansion

$$w = - \int_{V_i}^{V_f} p_{\text{ex}} dV$$

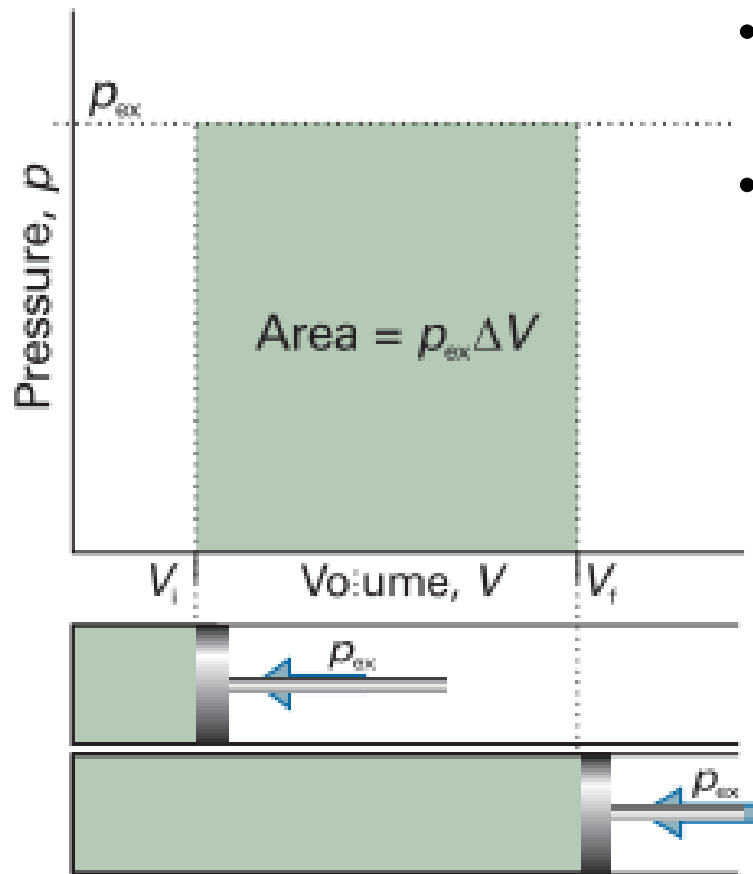
- expansion against zero opposing force,

$$P_{\text{ex}} = 0, \quad w = 0$$

- That is, **no work** is done when a system expands freely.
- Expansion of this kind occurs when a system expands into a vacuum.

Different types of expansion

(ii) Expansion against **constant pressure**



- The external pressure is **constant** throughout the expansion.
- For example, the piston may be pressed on by the atmosphere, which exerts the same pressure throughout the expansion.

$$w = - \int_{V_i}^{V_f} p_{\text{ex}} dV$$

$$w = -p_{\text{ex}} \int_{V_i}^{V_f} dV = -p_{\text{ex}}(V_f - V_i)$$

$$w = -p_{\text{ex}} \Delta V$$

The work done by a gas when it expands against a constant external pressure, P_{ex} is equal to the shaded area

Different types of expansion

(iii) Reversible expansion

$$w = - \int_{V_i}^{V_f} p_{\text{ex}} dV$$

- A reversible change in thermodynamics is a change that can be reversed by an *infinitesimal modification of a variable*.
- The key word ‘infinitesimal’ sharpens the everyday meaning of the word ‘reversible’ as something *that can change direction (reverse)*.
- To achieve reversible expansion we set p_{ex} equal to internal pressure (p) at each stage of the expansion.
- In practice, this equalization could be *achieved by gradually removing weights from the piston* so that the downward force due to the weights always matched the changing upward force due to the pressure of the gas.
- When we set $p_{\text{ex}} = p$ $w = -p dV$

Different types of expansion

$$w = - \int_{V_i}^{V_f} p_{\text{ex}} dV$$

(iv) Isothermal reversible expansion

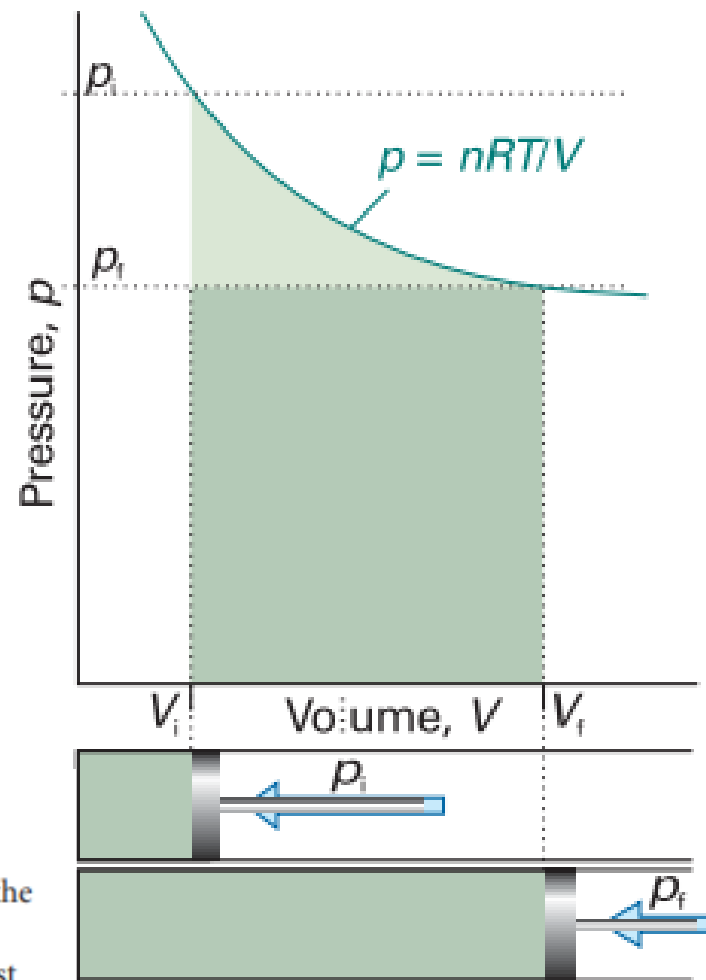
- Consider a perfect gas is expanding under constant temperature (Isothermal),

$$p = nRT/V$$

$$w = - \int_{V_i}^{V_f} p dV$$

$$w = -nRT \int_{V_i}^{V_f} \frac{dV}{V} = -nRT \ln \frac{V_f}{V_i}$$

Fig. 2.8 The work done by a perfect gas when it expands reversibly and isothermally is equal to the area under the isotherm $p = nRT/V$. The work done during the irreversible expansion against the same final pressure is equal to the rectangular area shown slightly darker. Note that the reversible work is greater than the irreversible work.



Work done under reversible and irreversible process

Whole area under the PV curve =
work done under **reversible process**

The dark rectangular area under
PV curve = work done under **irreversible process**

In irreversible process **compression or expansion is** drastic/quick. So more energy is lost as heat.

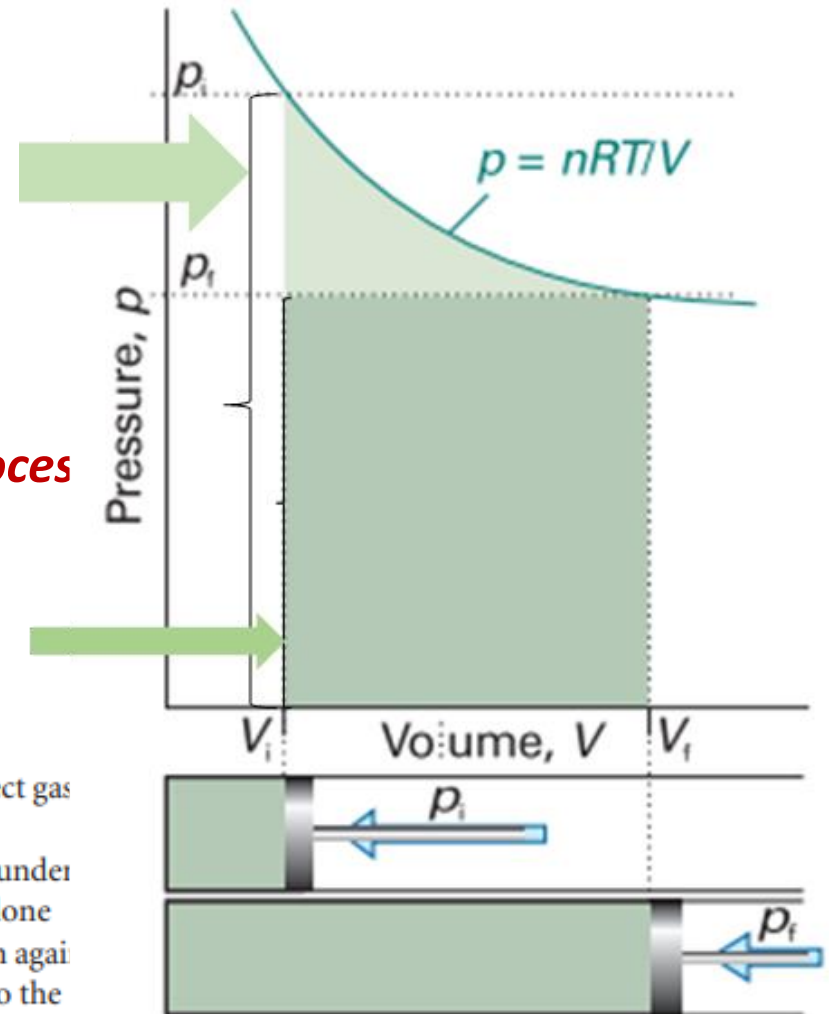


Fig. 2.8 The work done by a perfect gas when it expands reversibly and isothermally is equal to the area under isotherm $p = nRT/V$. The work done during the irreversible expansion against the same final pressure is equal to the rectangular area shown slightly darker. Note that the reversible work is greater than the irreversible work.

Summary for expansion work

- The work of free expansion is $w = 0$.
- The work of expansion against a constant external pressure is $w = -p_{\text{ex}}\Delta V$
- The work of isothermal reversible expansion of a perfect gas is $w = -nRT \ln(V_f/V_i)$

$$w = nRT \ln(V_i/V_f)$$

Example 2.1 *Calculating the work of gas production*

Calculate the work done when 50 g of iron reacts with hydrochloric acid in (a) a closed vessel of fixed volume, (b) an open beaker at 25°C.



Molar mass of Fe: 55.85 g mol⁻¹

Method to solve this problem:

We need to judge the **magnitude of the volume change** and then to decide how the process occurs.

If there is **no change in volume**, there is **no expansion work** however the process takes place.

If the system expands against a **constant external pressure**, the work can be calculated from eqn **$w = -p_{\text{ex}}\Delta V$** .

A general feature of processes in which a condensed phase changes into a gas is that the volume of the former may usually be neglected relative to that of the gas it forms.

Answer In (a) the volume cannot change, so no expansion work is done and $w = 0$. In (b) the gas drives back the atmosphere and therefore $w = -p_{\text{ex}} \Delta V$. We can neglect the initial volume because the final volume (after the production of gas) is so much larger and $\Delta V = V_f - V_i \approx V_f = nRT/p_{\text{ex}}$, where n is the amount of H_2 produced. Therefore,

$$w = -p_{\text{ex}} \Delta V \approx -p_{\text{ex}} \times \frac{nRT}{p_{\text{ex}}} = -nRT$$

According to our approximation $V_f \gg V_i$
 $\Delta V = V_f$ and $V_f = nRT/p_{\text{ex}}$

Because the reaction is $\text{Fe(s)} + 2 \text{HCl(aq)} \rightarrow \text{FeCl}_2(\text{aq}) + \text{H}_2(\text{g})$, we know that 1 mol H_2 is generated when 1 mol Fe is consumed, and n can be taken as the amount of Fe atoms that react. Because the molar mass of Fe is 55.85 g mol^{-1} , it follows that

$$\begin{aligned} w &\approx -\frac{50 \text{ g}}{55.85 \text{ g mol}^{-1}} \times (8.3145 \text{ J K}^{-1} \text{ mol}^{-1}) \times (298 \text{ K}) \\ &\approx -2.2 \text{ kJ} \end{aligned}$$

The system (the reaction mixture) does 2.2 kJ of work driving back the atmosphere. Note that (for this perfect gas system) the magnitude of the external pressure does not affect the final result: the lower the pressure, the larger the volume occupied by the gas, so the effects cancel.

Enthalpy (H) & Enthalpy Change (ΔH)

- Enthalpy is the amount of **heat** stored in the system under particular condition and is termed as heat content of the system.

$$H = U + PV$$

- Enthalpy change, $\Delta H = H_2 - H_1$

$$\Delta H = \Delta U + P\Delta V$$

Heat Capacity (C)

- Heat capacity is **the amount of heat required to change the temperature of a given amount of matter by 1°C.**
- The heat capacity of 1 gram of a substance is called its **specific heat capacity** (or specific heat),
- The heat capacity of 1 mole of a substance is called its **molar heat capacity**.

Specific Heats of Some Common Substances	
Substance	Specific Heat (J/goC)(J/goC)
Water (l)	4.18
Water (s)	2.06
Water (g)	1.87
Ammonia (g)	2.09
Ethanol (l)	2.44
Aluminum (s)	0.897
Carbon, graphite (s)	0.709

Heat Capacity (C)

water has a very **high specific heat** compared to most other substances.

Water is commonly used as a **coolant** for machinery because it is able to absorb large quantities of heat



Heat Capacity (C)

$$C = \frac{Q}{m d T}$$

$$\Delta U = q + w$$

$$C_m = \frac{Q}{d T} = \frac{d U - w}{d T}$$

$$\Delta U = q - P \Delta V$$

$$C_m = \frac{d U + p d V}{d T}$$

$$C_v = \left(\frac{d U}{d T} \right)_v$$

$$C_p - C_v = n R$$

$$C_m = \frac{d H}{d T}$$

$$C_p = \left(\frac{d H}{d T} \right)_p$$

$$\gamma = \frac{C_p}{C_v}$$

Adiabatic Changes

$$\Delta U = q + w$$

For **adiabatic process**, **q = 0**; as adiabatic wall does not allow the heat exchange.

$$\Delta U = w_{\text{ad}}$$

$$\Delta U = nC_V(T_f - T_i) = C_V\Delta T$$

C_V : **Heat capacity** of the gas at constant volume

$$w_{\text{ad}} = nC_V\Delta T \quad \text{work done under adiabatic condition}$$

Molecular Connection

$$w_{\text{ad}} = C_V \Delta T$$

The work done during an adiabatic expansion of a perfect gas

$$w_{\text{ad}} \propto \Delta T \text{ or } (T_f - T_i)$$

That is exactly what we expect on molecular grounds, *because the mean kinetic energy is proportional to T , so a change in internal energy arising from temperature alone is also expected to be proportional to ΔT .*

Molecular Connection

The work done during an adiabatic expansion of a perfect gas

$$W_{\text{ad}} \propto \Delta T = T_f - T_i$$

Higher the temperature raise, higher the rise of *mean kinetic energy of molecules in the system*.

At higher temperature, the internal energy of the system is high.

So, a change in internal energy arising from temperature alone is also expected to be proportional to ΔT and which is again proportional to W

The Initial and Final Temperature

$$w_{\text{ad}} = C_V \Delta T$$

$$T_f = T_i \left(\frac{V_i}{V_f} \right)^{1/c}$$

where $c = C_{V,m}/R$,

$C_{V,m}$ = constant molar volume heat capacity of the gas

$$V_i T_i^c = V_f T_f^c$$

This result is often summarized in the form

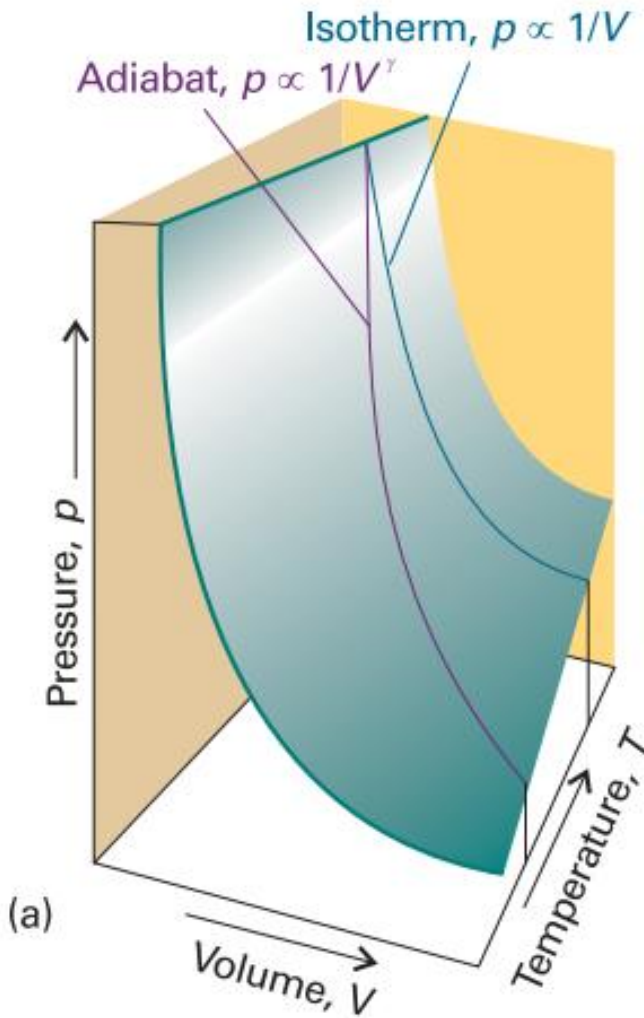
$$V T^c = \text{constant}$$

Initial and final **PV** Relationship

$$V T^\gamma = \text{constant}$$

$$p_f V_f^\gamma = p_i V_i^\gamma$$

$$\text{where } \gamma = C_{p,m}/C_{V,m}$$



The *curves of pressure versus volume for adiabatic change are known as **adiabats***, and one for a reversible path is illustrated in in the given figure.

We know that $\gamma > 1$, as $C_p > C_v$

So, an adiabat falls more steeply ($p \propto 1/V^\gamma$) than the corresponding isotherm ($p \propto 1/V$).

Illustration 2.5 *Work of adiabatic expansion*

Consider the adiabatic, reversible expansion of 0.020 mol Ar, initially at 25°C, from 0.50 dm³ to 1.00 dm³. The molar heat capacity of argon at constant volume is 12.48 J K⁻¹ mol⁻¹, so $c = 1.501$. Therefore, from eqn 2.28a,

$$T_f = (298 \text{ K}) \times \left(\frac{0.50 \text{ dm}^3}{1.00 \text{ dm}^3} \right)^{1/1.501} = 188 \text{ K}$$

It follows that $\Delta T = -110 \text{ K}$, and therefore, from eqn 2.27, that

$$w = \{(0.020 \text{ mol}) \times (12.48 \text{ J K}^{-1} \text{ mol}^{-1})\} \times (-110 \text{ K}) = -27 \text{ J}$$

Note that temperature change is independent of the amount of gas but the work is not.

Self-test 2.5 Calculate the final temperature, the work done, and the change of internal energy when ammonia is used in a reversible adiabatic expansion from 0.50 dm³ to 2.00 dm³, the other initial conditions being the same.

[195 K, -56 J, -56 J]

Illustration 2.6 *The pressure change accompanying adiabatic expansion*

When a sample of argon (for which $\gamma = \frac{5}{3}$) at 100 kPa expands reversibly and adiabatically to twice its initial volume the final pressure will be

$$p_f = \left(\frac{V_i}{V_f} \right)^\gamma p_i = \left(\frac{1}{2} \right)^{5/3} \times (100 \text{ kPa}) = 32 \text{ kPa}$$

For an isothermal doubling of volume, the final pressure would be 50 kPa.

For example

$$V_i = 50 \text{ dm}^3$$

$$V_f = 100 \text{ dm}^3$$

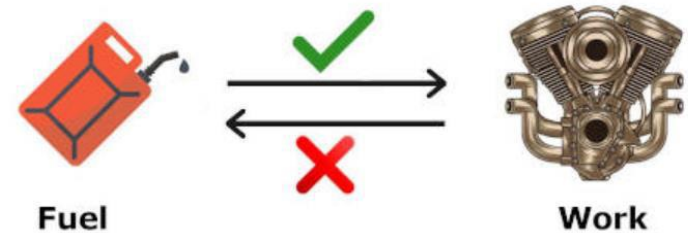
$$V_i/V_f = 50/100 = 1/2$$

First Law- Directionality of changes ?

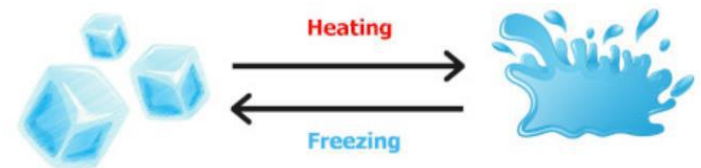
- ❑ When an ice cube melts in your hand, the first law tells that energy from your hand was transferred to the ice to change it to a liquid. If you could measure the work and heat involved in each case, you would find that **energy is conserved** as it is converted from one form to another.
- ❑ However, the first law **does not** help us make sense of the **direction** of the change.
- ❑ Why doesn't the pool of water in your cupped hand transfer the heat back to your hand and refreeze? Neither of these events would violate the first law—energy would still be conserved—but they never happen
- ❑ The first law by itself tells nothing about the direction of a spontaneous change, so we must look elsewhere for a way to predict that direction.

First Law- Directionality of changes ?

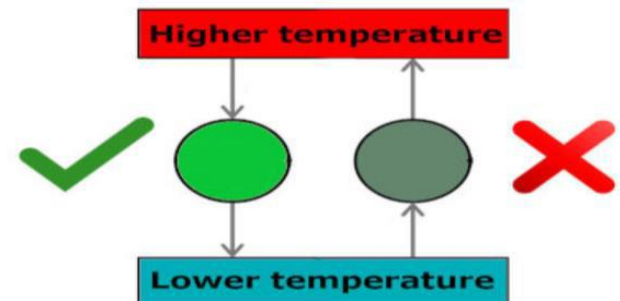
1st law of thermodynamics doesn't say anything about whether the process is possible or not



1st law of thermodynamics doesn't say whether the process will occur on its own or not



1st law of thermodynamics doesn't provide any sufficient condition for a certain process to take place



Limitation of 1st Law of Thermodynamics

- First law of thermodynamic is the law of “conservation of energy”
- It tell us the possibility of converting one form of energy into other form.
- But it does tell about the thermodynamically allowed direction of the conversion (or change).

Limitations of 1st Law of thermodynamics

- ❑ Heat flows from hot body to cold body, but NOT the *vice-versa*.
- ❑ Feasibility of a process (spontaneity of a process) 1st law is silent about it.
- ❑ All natural spontaneous processes are irreversible; 1st law is silent about it.
- ❑ All work can be transferred to heat, but NOT the *vice-versa*.

- Types of system
- The Zeroth Law
- Internal Energy (E)

$$\Delta E = q - w$$

- First Law of Thermodynamics
- Limitations of 1st Law of thermodynamics

- Enthalpy (H) & Enthalpy Change (ΔH)

$$\Delta H = \Delta E + P\Delta V$$

Second law of thermodynamics

“Some things happen naturally; some things don’t”

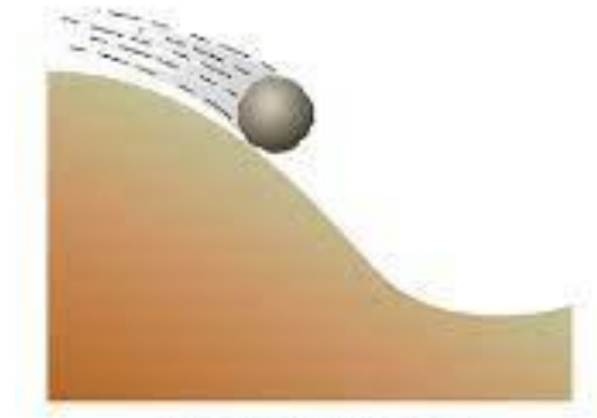
- The thing that occurs in a specific direction are called **spontaneous** change.
- For example **cooling of your hot coffee to its surrounding temperature.**
- The “spontaneous changes are always accompanied by a dispersal of energy” (***specifically thermal energy = heat***).

Kelvin Planck and Clausius Statements

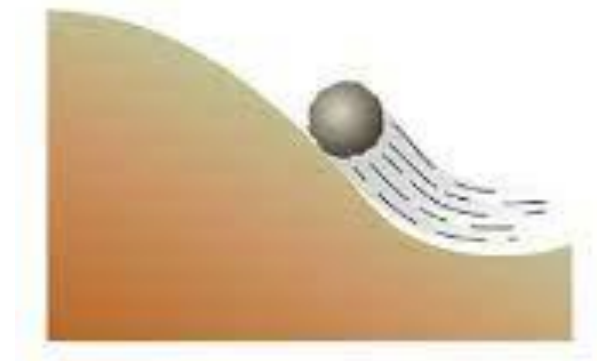
- The second law of thermodynamics “describes the **spontaneity** of a specific thermodynamic process with the help of various equivalent statements”
- Kelvin Planck’s and Clausius statements provide the formwork to understand the *dispersal of thermal energy as essential criteria of spontaneous events*.

Spontaneous Process

- Spontaneous processes are those that can proceed without any outside intervention.
- Processes that are spontaneous in one direction are nonspontaneous in the reverse direction.



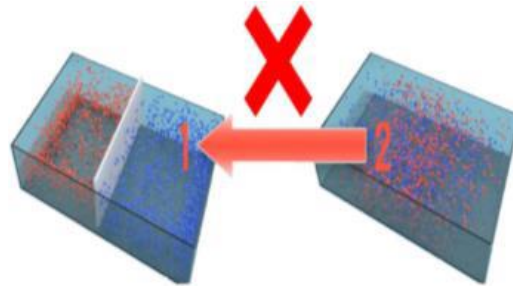
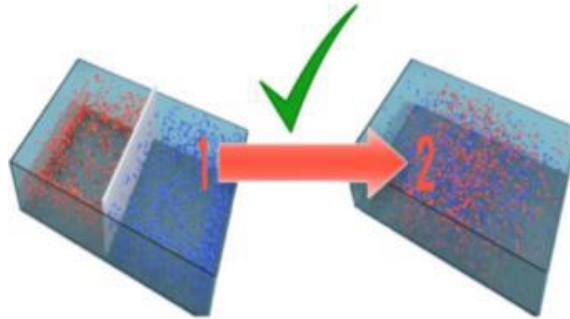
Spontaneous process



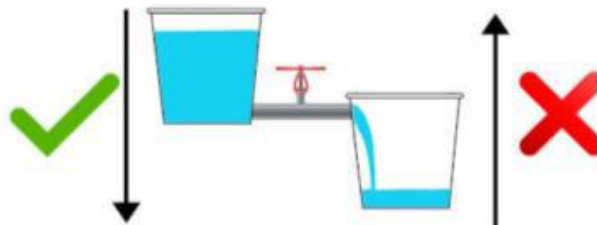
Nonspontaneous process

Second law of thermodynamics

examples/applications in everyday life



Water always flows from higher level to the lower level



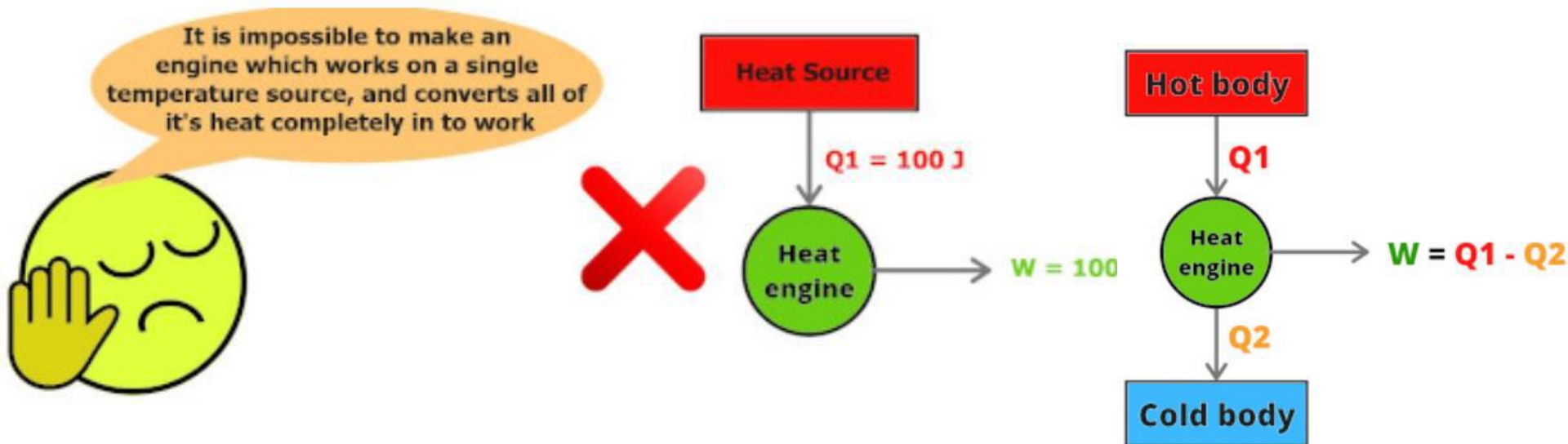
Water flows spontaneously from higher level to lower level



Second law of thermodynamics for Heat Engine

Kelvin Plank's Statements

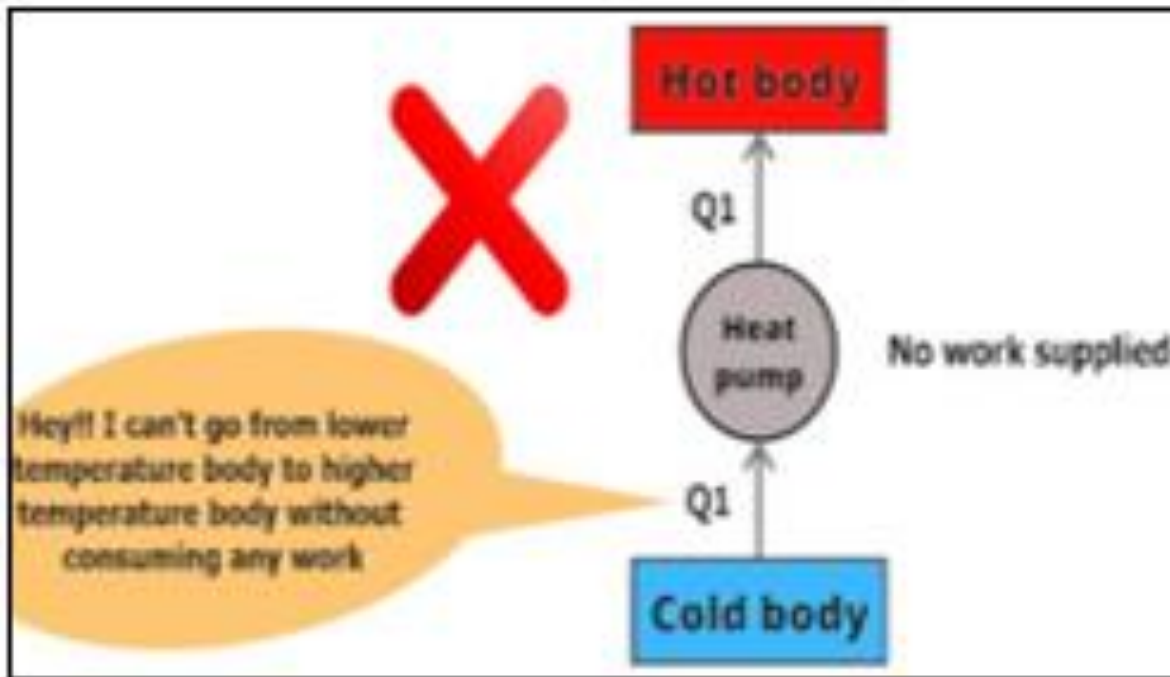
- It is impossible to make an engine which works on a single temperature source, and converts all of its heat completely into work.



Second law of thermodynamics for Heat Pump / Refrigerator

Clausius's Statements

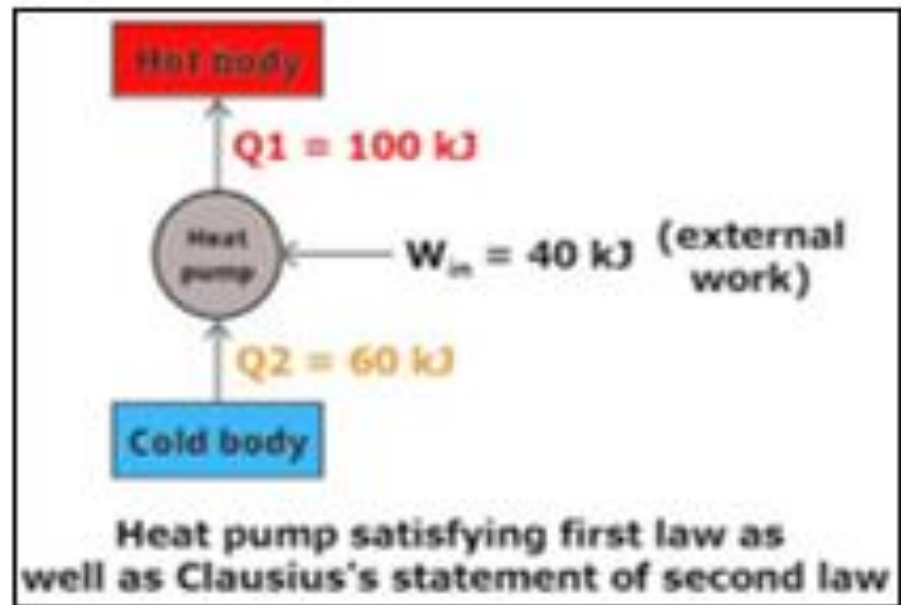
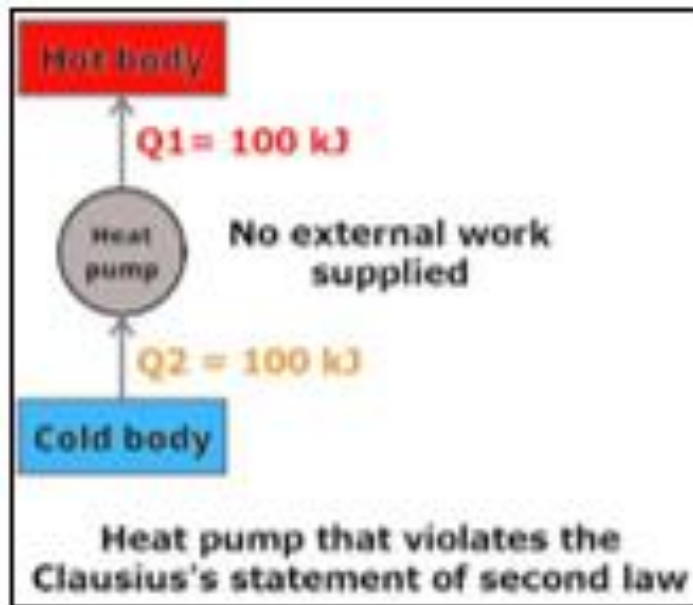
- It is impossible to construct a device (operating in a cycle) that can transfer heat from cold body to the hot body without absorbing any work.



Second law of thermodynamics for Heat Engine

Clausius's Statements

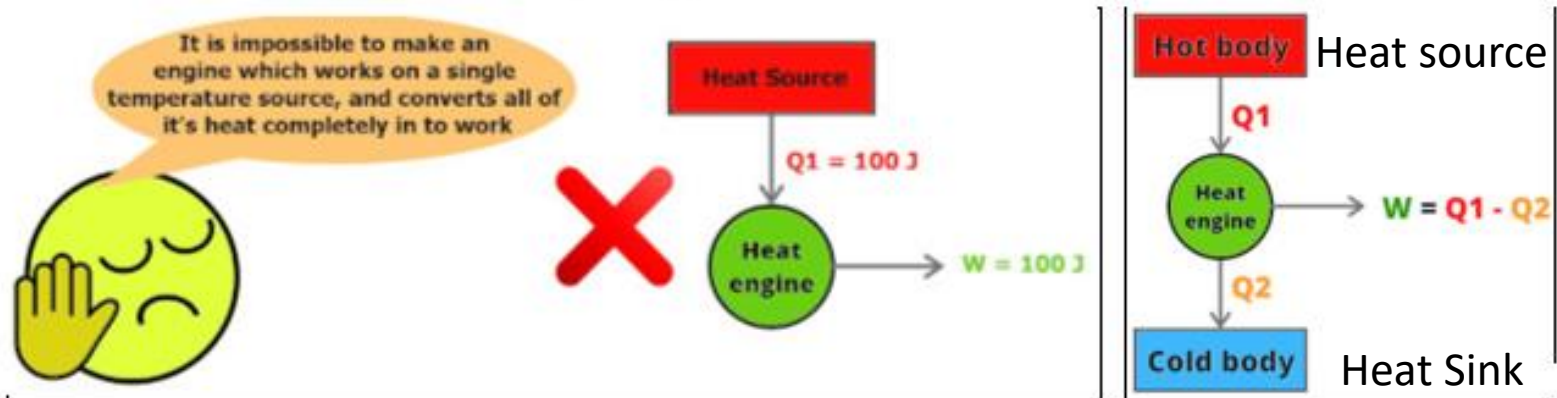
- It is impossible to construct a device (operating in a cycle) that can transfer heat from cold body to the hot body without absorbing any work.



Kelvin Plank's and Clausius's Statements

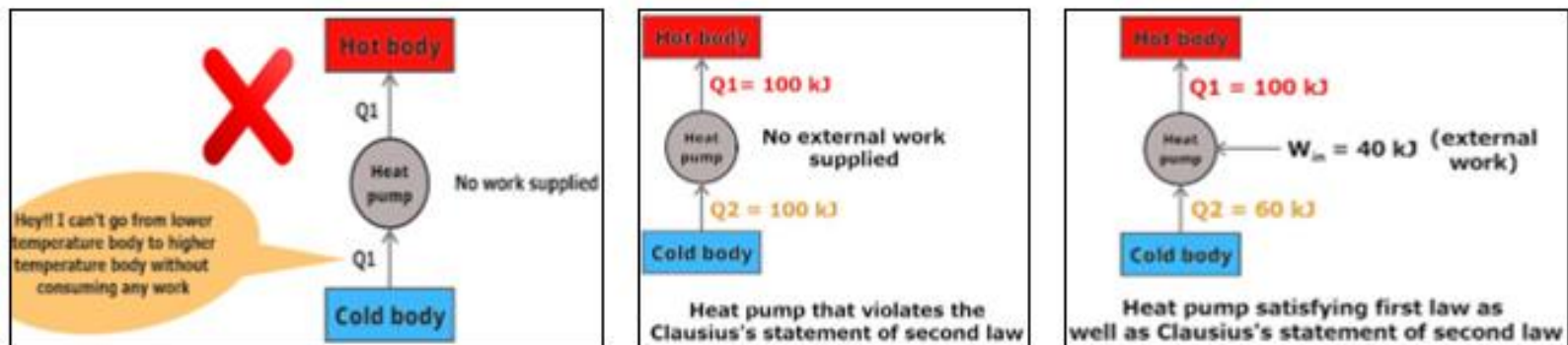
Second law of thermodynamics for heat engine (Kelvin Plank's statement):

It is impossible to make an engine which works on a single temperature source, and converts all of its heat completely in to work.



Second law of thermodynamics for heat pump / refrigerator (Clausius's statement):

It is impossible to construct a device (operating in a cycle) that can transfer heat from cold body to the hot body without absorbing any work.



Second Law and Entropy: Entropy (S)

- The Second Law of thermodynamics analyses these statements in terms of the entropy (**disorder/randomness**)
- Entropy is a state function denoted by **S**
- The entropy of an isolated system increases in the course of a spontaneous change:

$$\Delta S_{\text{tot}} > 0$$

where S_{tot} is the total entropy of the system and its surroundings.

ENTROPY (S)

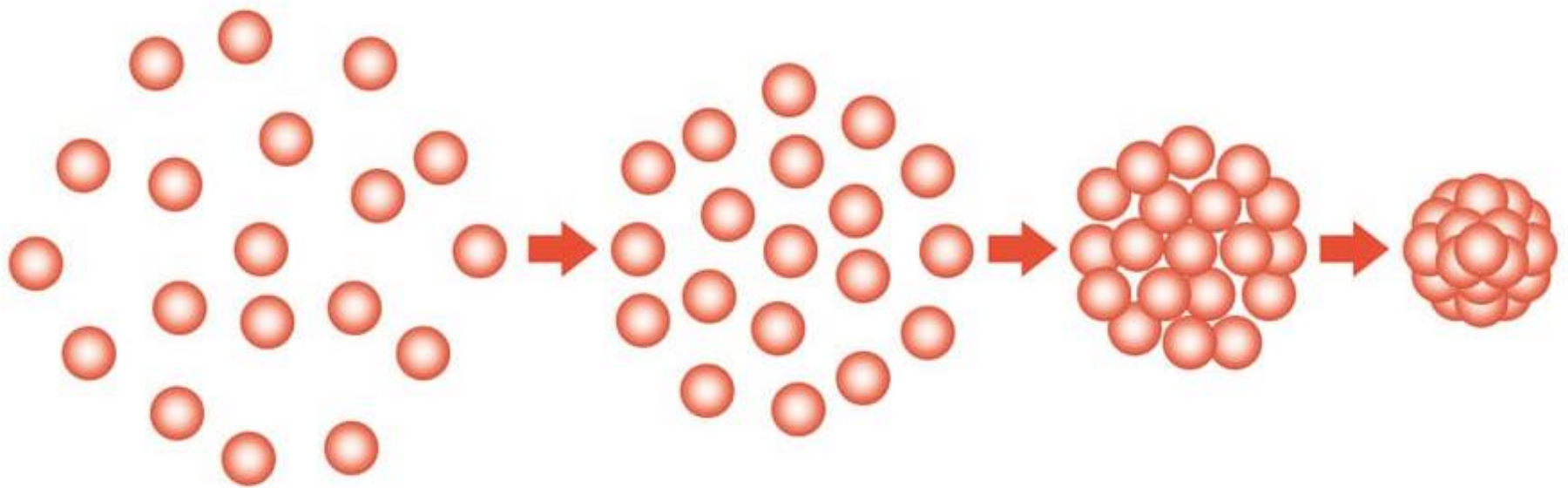
- Entropy can be thought of as a measure of the **randomness** of a system.

$$S_{\text{gas}} > S_{\text{liquid}} > S_{\text{solid}}$$

$$\Delta S = \frac{q_{\text{rev}}}{T}$$

$$\Delta S = S_{\text{final}} - S_{\text{initial}}$$





High Randomness,
High Entropy, High Disorder

Low Randomness,
Low Entropy, Low Disorder

Second Law of Thermodynamics

- The entropy of the universe does not change for reversible processes and increases for spontaneous processes.

$$\Delta S_{univ} = \Delta S_{system} + \Delta S_{surroundings} > 0$$

Where,

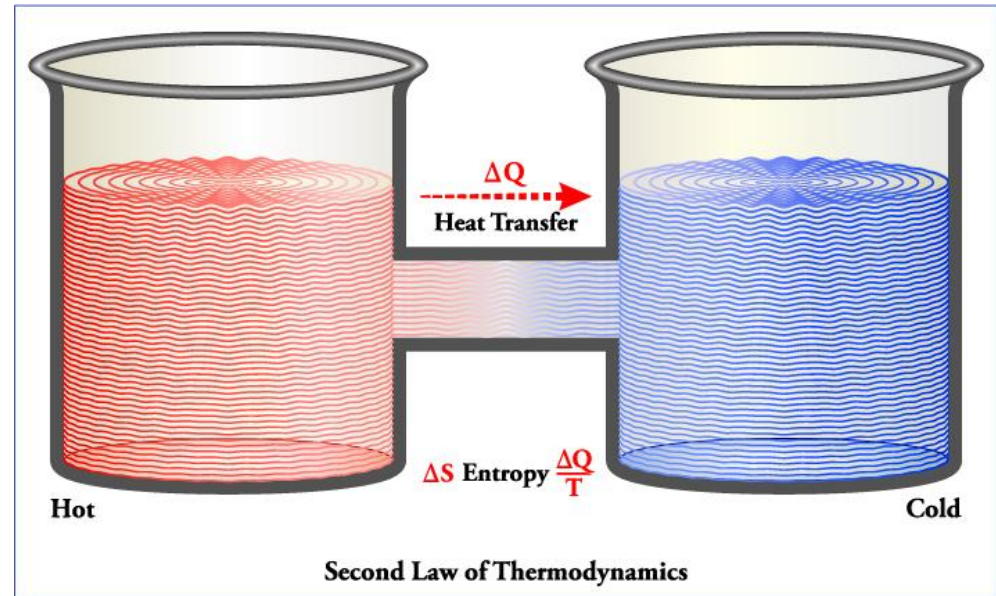
$\Delta S_{total} > 0$ (Irreversible (real, spontaneous):

$\Delta S_{total} = 0$ (equilibrium) Reversible (ideal):

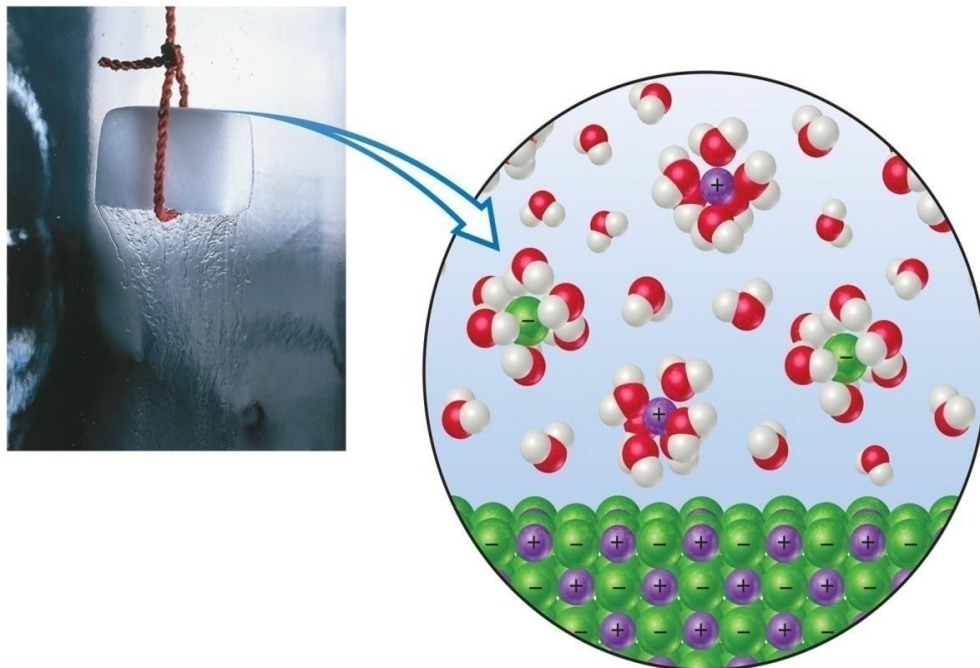
$\Delta S_{total} < 0$ non-spontaneous

The Second Law of Thermodynamics

In any **spontaneous process**, the total entropy change of the universe is positive, ($\Delta s_u > 0$)



$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$$



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Thermodynamic Irreversibility

“Spontaneous process are thermodynamically irreversible process”

Examples

1. Transfer of heat from the hot coffee to it's cold surrounding
2. The free expansion of gases into the vacuum

These process must be accompanied by an increase in total entropy.

$$\Delta S_{\text{tot}} = \Delta S_{\text{sys}} + \Delta S_{\text{sur}} \gg 0$$

Thermodynamic definition of entropy

The thermodynamic definition of entropy is based on the expression

$$dS = \frac{dq_{rev}}{T}$$

For a measurable change between two states i and f this expression integrates to

$$\Delta S = \int_i^f \frac{dq_{rev}}{T}$$

Important Points About Entropy:

- ❑ Absolute entropy can NOT be measured! Change in entropy can be measured.
- ❑ Entropy for a **spontaneous** reaction is **+ve** and *vice-versa*.
- ❑ Entropy for an **equilibrium** reaction is **ZERO**.
- ❑ Entropy is a state function; $\Delta S = S_{\text{final}} - S_{\text{initial}}$
- ❑ Entropy is an **extensive property**!
- ❑ Unit of entropy: **J/mol.K**
- ❑ Entropy of the universe is always **INCREASING**!

Calculating entropy change for isothermal process

Example 3.1 *Calculating the entropy change for the isothermal expansion of a perfect gas*

Calculate the entropy change of a sample of perfect gas when it expands isothermally from a volume V_i to a volume V_f .

Method The definition of entropy instructs us to find the energy supplied as heat for a reversible path between the stated initial and final states regardless of the actual manner in which the process takes place. A simplification is that the expansion is isothermal, so the temperature is a constant and may be taken outside the integral in eqn 3.2. The energy absorbed as heat during a reversible isothermal expansion of a perfect gas can be calculated from $\Delta U = q + w$ and $\Delta U = 0$, which implies that $q = -w$ in general and therefore that $q_{\text{rev}} = -w_{\text{rev}}$ for a reversible change. The work of reversible isothermal expansion was calculated in Section 2.3.

If the change is isothermal, T is constant, then change in **internal energy is zero**

Answer Because the temperature is constant, eqn 3.2 becomes

$$\Delta S = \frac{1}{T} \int_i^f dq_{\text{rev}} = \frac{q_{\text{rev}}}{T}$$

From eqn 2.11, we know that

$$q_{\text{rev}} = -w_{\text{rev}} = nRT \ln \frac{V_f}{V_i}$$

It follows that

$$\Delta S = nR \ln \frac{V_f}{V_i}$$

As an illustration of this formula, when the volume occupied by 1.00 mol of any perfect gas molecules is doubled at any constant temperature, $V_f/V_i = 2$ and

$$\Delta S = (1.00 \text{ mol}) \times (8.3145 \text{ J K}^{-1} \text{ mol}^{-1}) \times \ln 2 = +5.76 \text{ J K}^{-1}$$

One mole of gas has volume $X = 50 \text{ cm}^3$),

Upon isothermal reversible expansion the volume becomes doubled, that means $2X$

Entropy change for **adiabatic process**

The easiest pathway for which to calculate entropy changes is an adiabatic pathway.

Since **$dq = 0$** for an adiabatic change,

then **$dS = 0$** as well.

Entropy change and **heat** (q_{rev})

- The entropy change (ΔS) is related to heat flow (q_{rev}) by **$\Delta S = q_{\text{rev}}/T$**
- At **constant pressure**, $q_{\text{rev}} = nC_p\Delta T = \text{Enthalpy change } (\Delta H)$
- At **constant volume**, $q_{\text{rev}} = nC_v\Delta T = \text{Internal energy change } (\Delta U)$

where **n** is the number of moles of substance present

Solution to the integrals

- The change in entropy for a substance whose temperature changes from T_1 to T_2 is as follows:

$$\Delta S = nC_p \ln \frac{T_2}{T_1} \quad (\text{constant pressure})$$

$$\Delta S = nC_v \ln \frac{T_2}{T_1} \quad (\text{constant volume})$$

Entropy change in the surrounding: Upon heat transfer and work

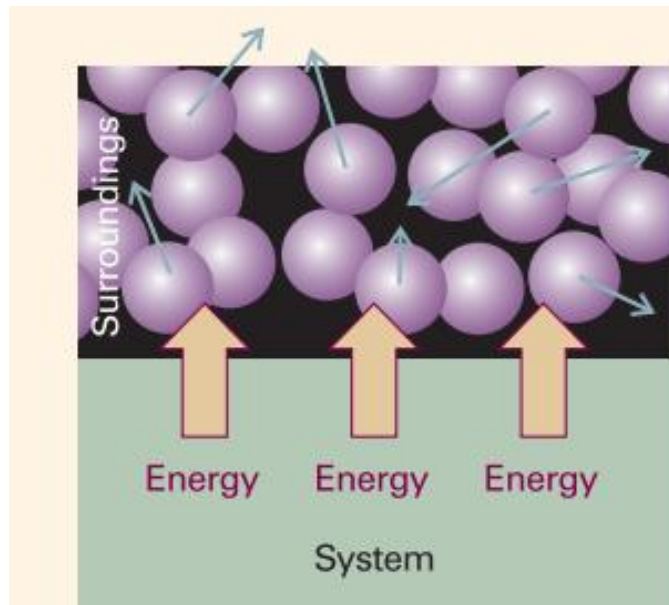


Fig. 2.3 When energy is transferred to the surroundings as heat, the transfer stimulates random motion of the atoms in the surroundings. Transfer of energy from the surroundings to the system makes use of random motion (thermal motion) in the surroundings.

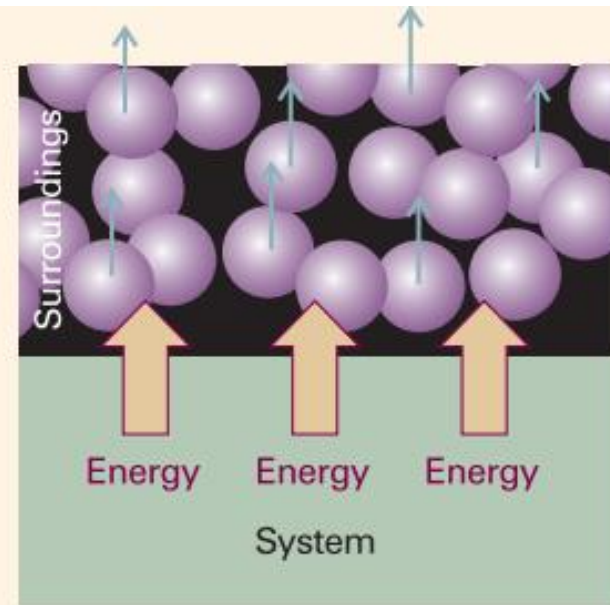


Fig. 2.4 When a system does work, it stimulates orderly motion in the surroundings. For instance, the atoms shown here may be part of a weight that is being raised. The ordered motion of the atoms in a falling weight does work on the system.

Upon release of heat from system, entropy of surrounding increases.
Upon system does work, the entropy of surrounding decreases.

Free energy change (ΔG)

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$$

- The Gibbs free energy change (**G**) is a measure of the **spontaneity** of a process and of the **useful energy** available from it.

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$$

$$\text{At constant pressure, } \Delta S_{\text{surr}} = -\frac{\Delta H_{\text{sys}}}{T}$$

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} - \frac{\Delta H_{\text{sys}}}{T}$$

$$-T\Delta S_{\text{univ}} = \Delta H_{\text{sys}} - T\Delta S_{\text{sys}}$$

$$\Delta G_{\text{sys}} = \Delta H_{\text{sys}} - T\Delta S_{\text{sys}}$$

Free energy change (ΔG)

$$\Delta G_{\text{sys}} = \Delta H_{\text{sys}} - T\Delta S_{\text{sys}}$$

Combining this equation with the previous one shows that

$$-T\Delta S_{\text{univ}} = \Delta H_{\text{sys}} - T\Delta S_{\text{sys}} = \Delta G_{\text{sys}}$$

*The **sign** of ΔG tells if a reaction is spontaneous.* The second law dictates

- $\Delta S_{\text{univ}} > 0$ for a spontaneous process
- $\Delta S_{\text{univ}} < 0$ for a nonspontaneous process
- $\Delta S_{\text{univ}} = 0$ for a process at equilibrium

Of course, absolute temperature is always positive, so

$$T\Delta S_{\text{univ}} > 0 \quad \text{or} \quad -T\Delta S_{\text{univ}} < 0 \text{ for a spontaneous process}$$

Because $\Delta G = -T\Delta S_{\text{univ}}$, we know that

- $\Delta G < 0$ for a spontaneous process
- $\Delta G > 0$ for a nonspontaneous process
- $\Delta G = 0$ for a process at equilibrium

Spontaneity of a reaction

$$\Delta G_{\text{sys}} = \Delta H_{\text{sys}} - T\Delta S_{\text{sys}}$$

The four possible combinations for the signs of ΔH and ΔS

Sign of ΔH	Sign of ΔS	Implications for Spontaneity
–	+	Spontaneous at all temperatures
+	–	Never spontaneous
–	–	Spontaneous only at low temperatures
+	+	Spontaneous only at high temperatures

Heat engines and the second law of thermodynamics

“A heat engine is a device that converts internal energy to mechanical energy”

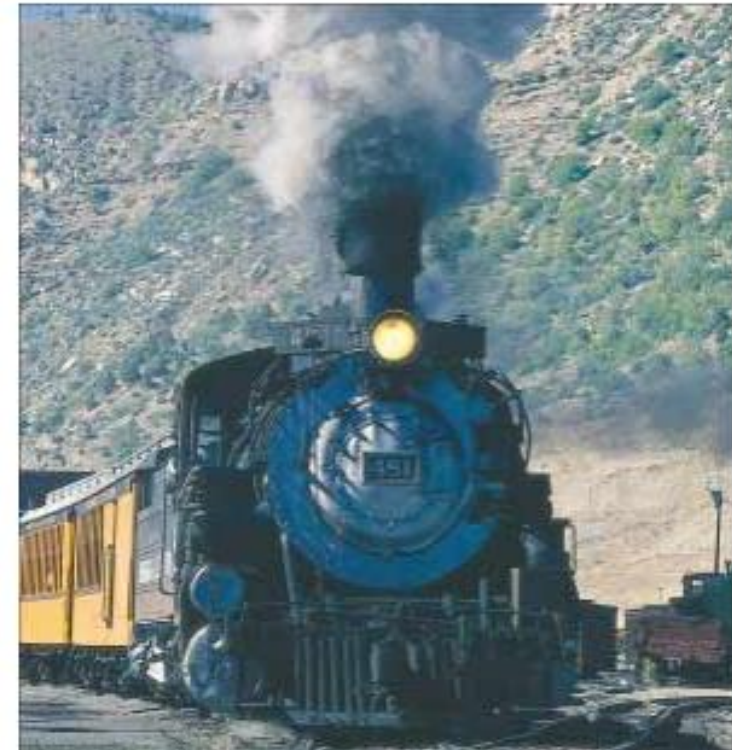
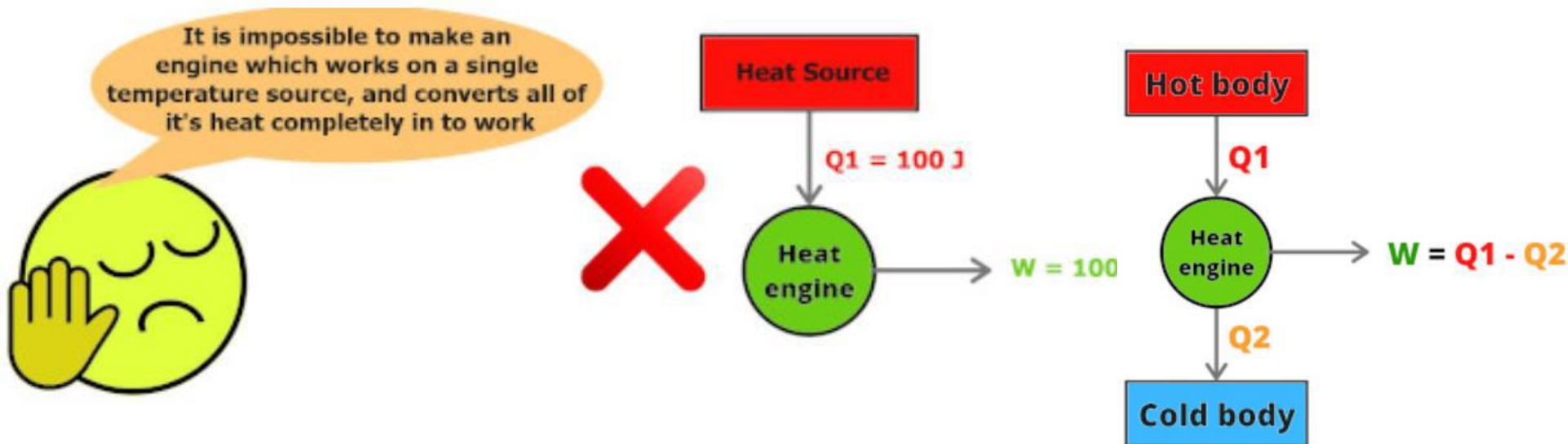


Figure 22.1 This steam-driven locomotive runs from Durango to Silverton, Colorado. It obtains its energy by burning wood or coal. The generated energy vaporizes water into steam, which powers the locomotive. (This locomotive must take on water from tanks located along the route to replace steam lost through the funnel.) Modern locomotives use diesel fuel instead of wood or coal. Whether old-fashioned or modern, such locomotives are heat engines, which extract energy from a burning fuel and convert a fraction of it to mechanical energy.

Heat engines and the second law of thermodynamics

Kelvin Plank's Statements

- It is impossible to make an engine which works on a single temperature source, and converts all of its heat completely into work.



Kelvin Planck's Statement

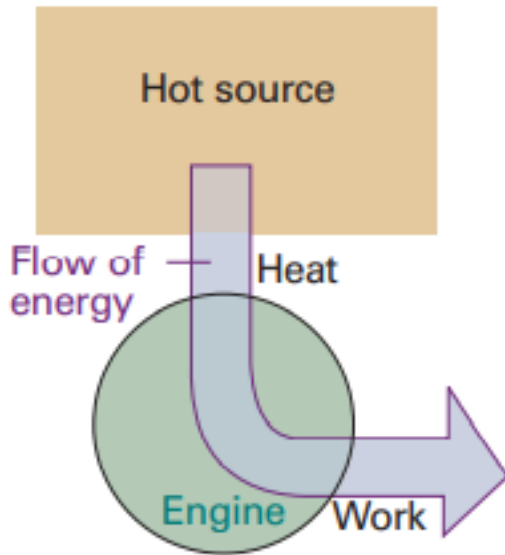


Fig. 3.1 The Kelvin statement of the Second Law denies the possibility of the process illustrated here, in which heat is changed completely into work, there being no other change. The process is not in conflict with the First Law because energy is conserved.

No process is possible in which the sole result is the absorption of heat from a reservoir and its complete conversion into work.

“Work can’t be extracted from a single heat reservoir”

A heat engine needs two reservoirs, a **heat reservoir** and a **heat sink**

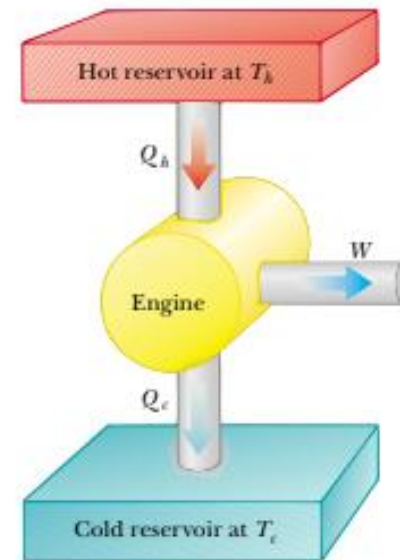


Figure 22.2 Schematic representation of a heat engine. The engine absorbs energy Q_h from the hot reservoir, expels energy Q_c to the cold reservoir, and does work W .

Thermal **efficiency** of a heat engine

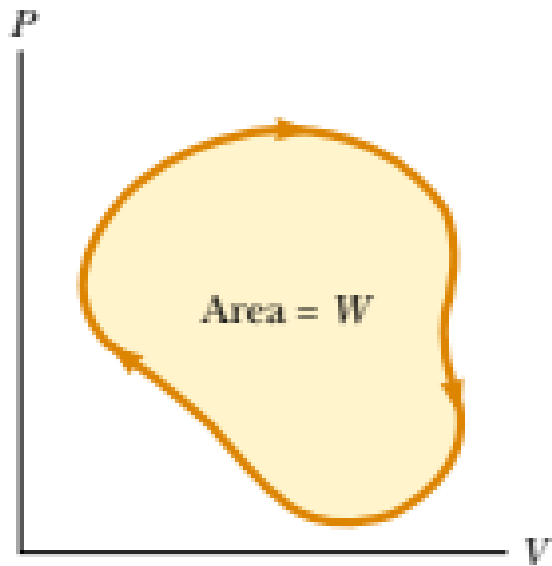


Figure 22.3 *PV* diagram for an arbitrary cyclic process. The value of the net work done equals the area enclosed by the curve.

The net work done in a **cyclic process** is the area enclosed by the curve representing the process on a *PV* diagram

The **thermal efficiency** (e) of a heat engine is defined as the ratio of the **net work done** by the engine during one cycle **to the energy** absorbed at the higher temperature during the cycle.

$$e = \frac{W}{Q_h} = \frac{Q_h - Q_c}{Q_h} = 1 - \frac{Q_c}{Q_h}$$

Thermal **efficiency** of a heat engine

$$e = \frac{W}{Q_h} = \frac{Q_h - Q_c}{Q_h} = 1 - \frac{Q_c}{Q_h}$$

A heat engine has 100% efficiency (**$e = 1$**), if and only if **$Q_c = 0$** .

It is impossible extract work from a heat source without transferring heat to sink (Kelvin Planck's statement), hence, **thermal efficiency of a engine is always $< 100\%$**

For example, a good automobile engine has an efficiency of about **20%**, and diesel engines have efficiencies ranging from **35% to 40%**.

THE CARNOT ENGINE

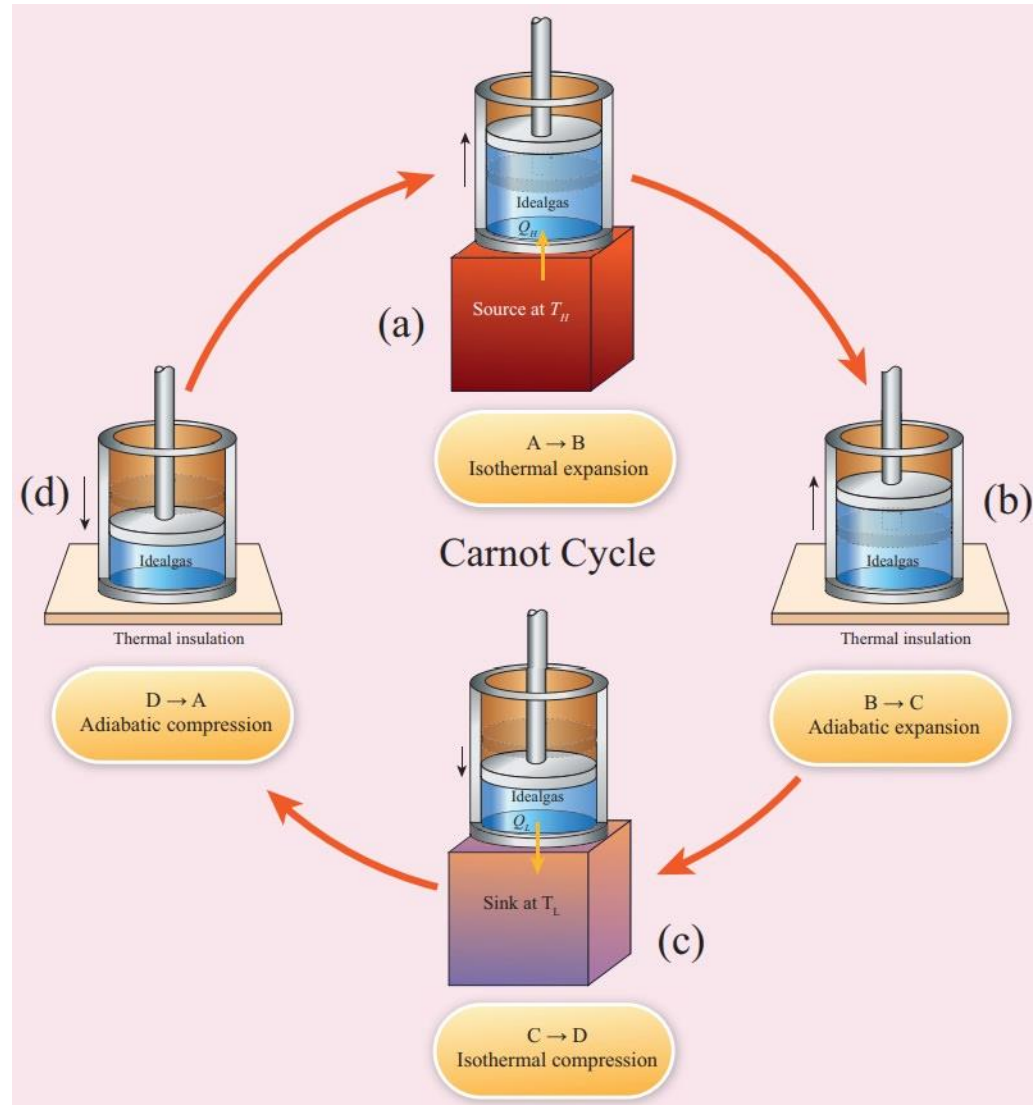
- In 1824 a French engineer named **Sadi Carnot** described a theoretical engine, now called a Carnot engine, that is of great importance from both practical and theoretical viewpoints.

- ***Carnot theorem***

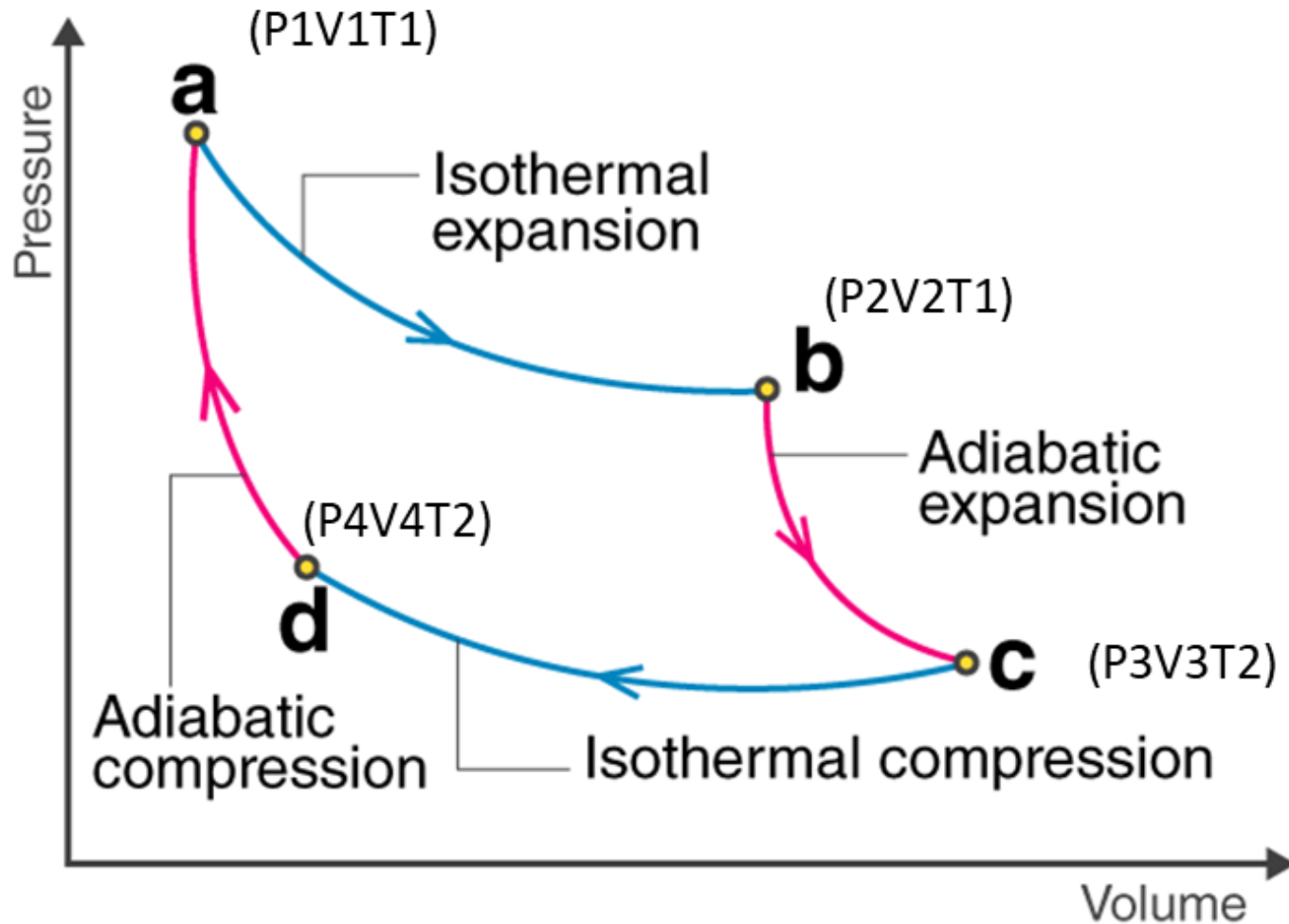
“No real heat engine operating between two energy reservoirs can be more efficient than a Carnot engine operating between the same two reservoirs”

Carnot Cycle

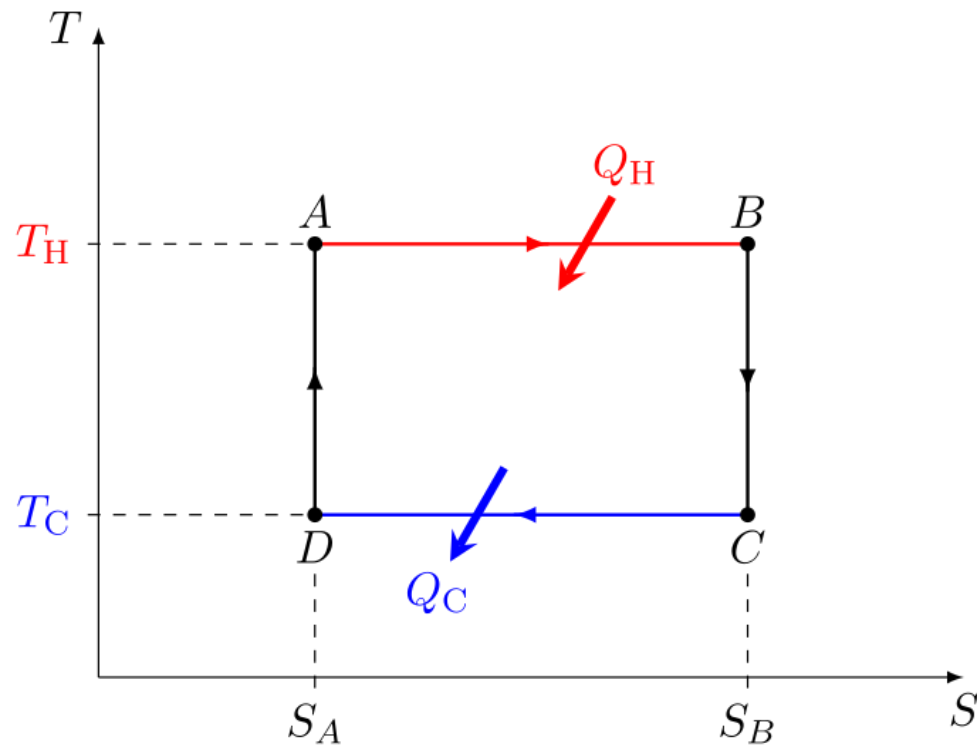
1. Isothermal expansion
2. Adiabatic expansion
3. Isothermal compression
4. Adiabatic compression



Carnot Cycle (P-V diagram)



Carnot cycle (T-S diagram)



Cyclic Process –Carnot Engine

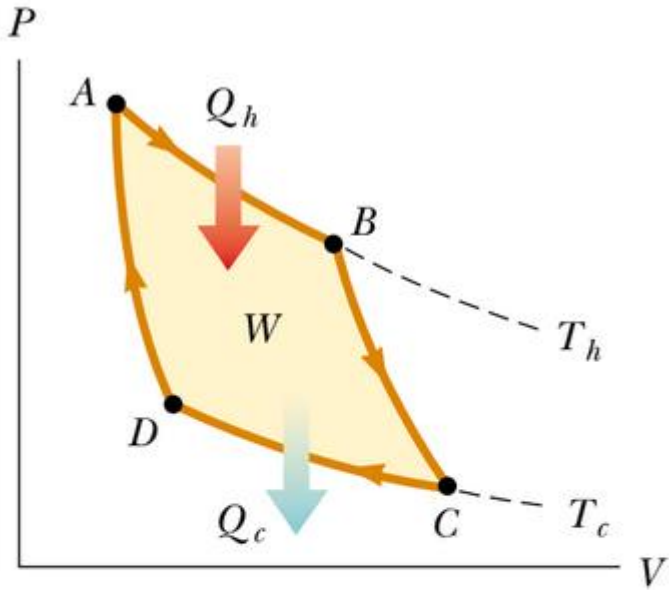


Figure 22.10 PV diagram for the Carnot cycle. The net work done, W , equals the net energy received in one cycle, $Q_h - Q_c$. Note that $\Delta E_{\text{int}} = 0$ for the cycle.

- A to B** – Isothermal expansion
- B to C** – Adiabatic expansion
- C to D** – Isothermal compression
- D to A** – Adiabatic compression

During **isothermal expansion** Q_h enters the engine (from source)

During **isothermal compression** Q_c enters the sink (from engine)

During **adiabatic expansion**,
Temperature drops from T_h to T_c

During **adiabatic compression**,
temperature of the system
rise from T_c to T_h

For a **cyclic reversible** path - Carnot Cycle

$$\Delta S = \int_i^f dS = \int_i^f \frac{dQ_r}{T} \quad (\text{reversible path})$$

$$\Delta S = \overset{0}{dS_{\text{adia}} (\text{expansion})} + \overset{0}{dS_{\text{adia}} (\text{compression})} + dS_{\text{iso}} (\text{expansion}) + dS_{\text{iso}} (\text{compression})$$

dQ for two adiabatic change in the **cycle** is zero, so **$dS = 0$**

For two isothermal process (expansion and compression) in the cycle

$$\Delta S = \frac{Q_h}{T_h} - \frac{Q_c}{T_c}$$

For Carnot engine

$$\Delta S = 0$$

$$\frac{Q_c}{Q_h} = \frac{T_c}{T_h}$$

$$\frac{Q_c}{T_c} = \frac{Q_h}{T_h}$$

Entropy change for Carnot cycle

$$\oint \frac{dQ_r}{T} = 0$$

where the symbol $\oint$ indicates that the integration is over a closed path.

The change in entropy for a Carnot cycle is zero

$\Delta S = 0$ for any reversible cycle

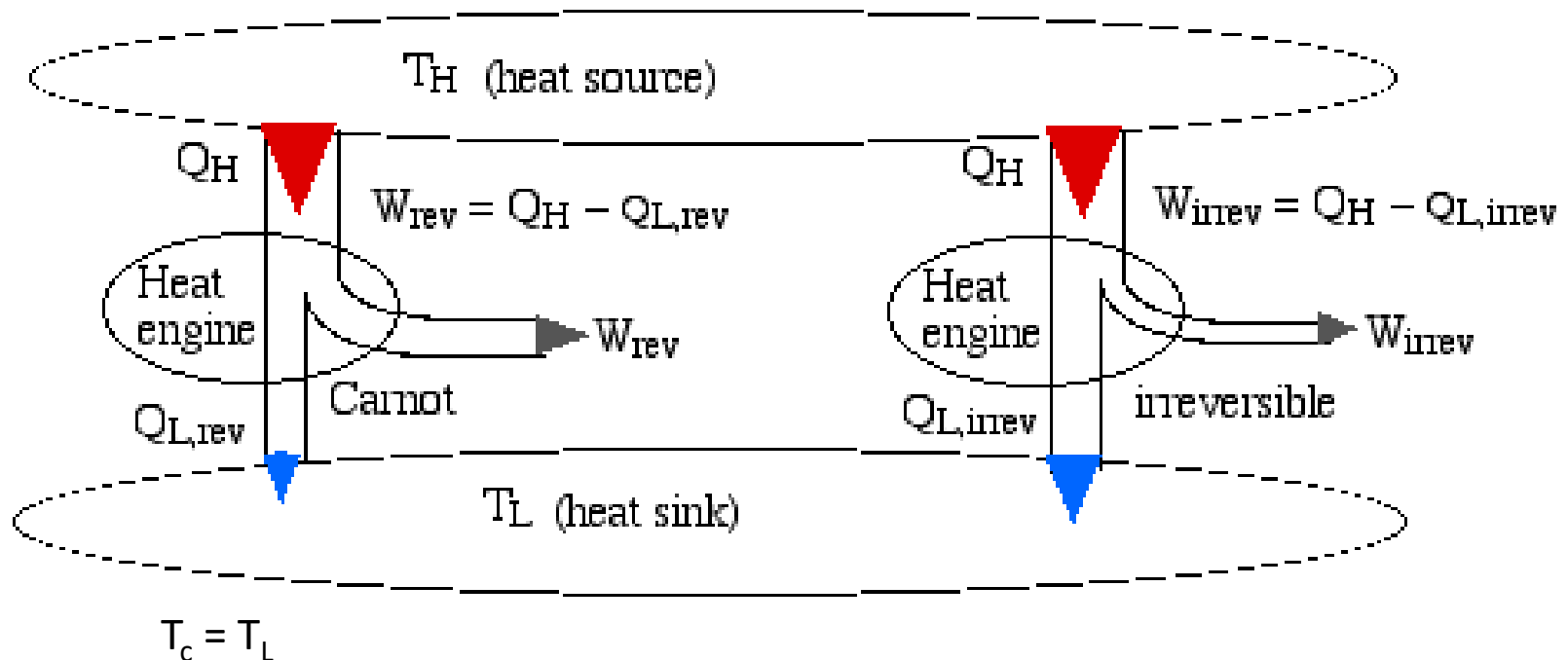
Carnot engine vs Real heat engine

- Practically, *the efficiency of real heat engine cycle is less than that of an ideal engine cycle (Carnot cycle).*
- The difference between the two cycle is that the expansion and compression process in Carnot engine is assumed to be reversible
Whereas in **real engine** these process occur in **irreversible** manner

*Work done under any irreversible process is less than that of the work done under a reversible process, because **less energy is lost as heat in a reversible process.***

Carnot engine vs Real heat engine

Assume that the Carnot engine and real heat engine draw the same amount of heat (Q_H) from the same heat source



In **reversible process**,
change of variable is infinitesimal
The heat lose (Q_L) to the sink is smaller

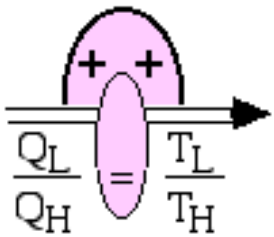
In **irreversible process**,
change in variable is abrupt(quick)
The heat lose (Q_L) to sink is larger

$$Q_{L, irrev} > Q_{L, rev}$$

Clausius Inequality

For Carnot Cycle

$$\oint \frac{\delta Q}{T} = \frac{Q_H}{T_H} - \frac{Q_{L,rev}}{T_L}$$



$$\frac{Q_{L,rev}}{T_L} = \frac{Q_H}{T_H}$$

For reversible case

$$dq/T = 0 = ds$$

For irreversible case

$$dq/T < 0$$

$$ds_{sys} < 0$$

reversible machines

$$\oint \frac{\delta Q}{T} = 0$$

For real engine cycle

$$Q_{L,irrev} > Q_{L,rev}$$

$$Q_{diff} = Q_{L,irrev} - Q_{L,rev}$$

$$\frac{Q_{L,irrev}}{T_L} = \left(\frac{Q_{L,rev}}{T_L} + \frac{Q_{diff}}{T_L} \right)$$

$$\oint \frac{\delta Q}{T} = \frac{Q_H}{T_H} - \left(\frac{Q_{L,rev}}{T_L} + \frac{Q_{diff}}{T_L} \right) < 0$$

The Clausius Inequality: $\oint \frac{\delta Q}{T} \leq 0$

= reversible process
< irreversible process

Alternatively, Clausius inequality is $dS \geq dQ/T$ (entropy change in sys + sur)

Total entropy change is positive for a spontaneous (irreversible change)

Carnot engine vs Real heat engine

For reversible cycles
(Carnot)

$$\oint \left(\frac{\delta Q}{T} \right) = 0$$

For irreversible cycles
(Real heat engine)

$$\oint \left(\frac{\delta Q}{T} \right) < 0$$

EXAMPLE 22.1 The Efficiency of an Engine

Find the efficiency of a heat engine that absorbs 2 000 J of energy from a hot reservoir and exhausts 1 500 J to a cold reservoir.

$$Q_c/Q_h = T_c/T_h \qquad e = 1 - \frac{Q_c}{Q_h} = 1 - \frac{1\,500\text{ J}}{2\,000\text{ J}} = 0.25, \text{ or } 25\%$$

The temperature must be used in Kelvin scale to solve this kind of problems, if T_c and T_h are given.

A certain power station operates with superheated steam at 300°C ($T_h = 573\text{ K}$) and discharges the waste heat into the environment at 20°C ($T_c = 293\text{ K}$). The theoretical efficiency is therefore

$$\eta = 1 - \frac{293\text{ K}}{573\text{ K}} = 0.489$$

or 48.9 per cent. In practice, there are other losses due to mechanical friction and the fact that the turbines do not operate reversibly.

The highest theoretical efficiency of a certain engine is 30%. If this engine uses the atmosphere, which has a temperature of 300 K, as its cold reservoir, what is the temperature of its hot reservoir?

$$e_C = 1 - \frac{T_c}{T_h}$$

$$T_h = \frac{T_c}{1 - e_C} = \frac{300 \text{ K}}{1 - 0.30} = 430 \text{ K}$$

Example: Quiz 18.2 Three engines operate between reservoirs separated in temperature by 300 K. The reservoir temperatures are as follows:

Engine A: $T_{hot} = 1000$ K, $T_{cold} = 700$ K

Engine B: $T_{hot} = 800$ K, $T_{cold} = 500$ K

Engine C: $T_{hot} = 600$ K, $T_{cold} = 300$ K

Rank the engines in terms of theoretically possible efficiencies, from lowest to highest.

because the efficiency $e = 1 - T_{cold}/T_{hot}$. So

Engine A: $T_{cold}/T_{hot} = 0.7$, $e = 0.3$, least efficient.

Engine B: $T_{cold}/T_{hot} = 0.625$, $e = 0.375$

Engine C: $T_{cold}/T_{hot} = 0.5$, $e = 0.5$, most efficient

While lower T_c provides higher theoretical efficiency, higher T_h is practically achievable.

Heat Pumps & Refrigerators

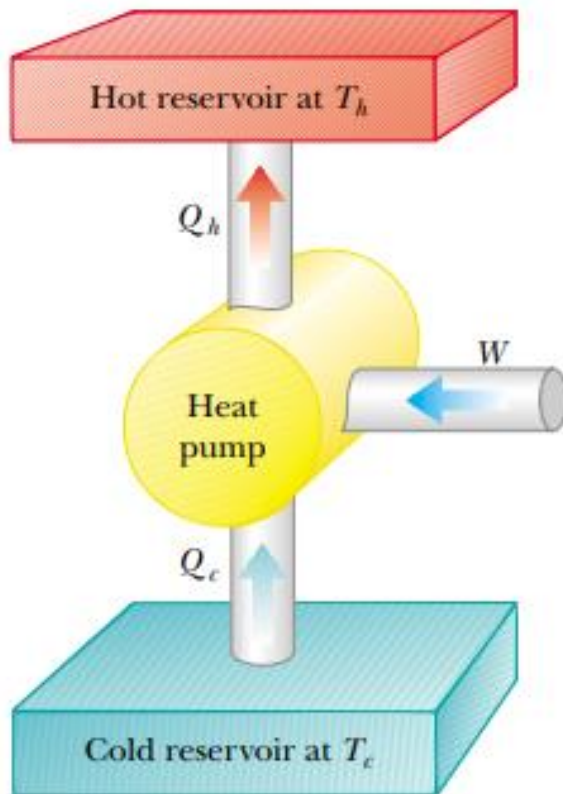
- **Heat pump** is a mechanical device that moves thermal energy from a region at lower temperature to a region at higher temperature.
- Heat pumps have long been used for **cooling** homes and buildings, and they are now becoming increasingly popular for **heating** them as well.

According to the **Clausius statement**,
*the pumping of heat from cold
reservoir to hot reservoir requires
external work (example electrical work)*



Heat Pumps & Refrigerators

Heating and cooling mode utilizes same principle



Theoretically, *a Carnot-cycle heat engine run in reverse cycle constitutes the most effective heat pump possible.*

Figure 22.15 Schematic diagram of a heat pump, which absorbs energy Q_c from a cold reservoir and expels energy Q_h to a hot reservoir. Note that this diagram is the same as that for the refrigerator shown in Figure 22.5.

Effectiveness of heat pump

- The effectiveness of a heat pump is described in terms of a number called the ***Coefficient of Performance*** (COP).
- In the **heating mode**, the COP is defined as the ratio of the energy transferred to the hot reservoir to the work required to transfer that energy:

$$\begin{aligned}\text{COP}_C(\text{heating mode}) &= \frac{Q_h}{W} \\ &= \frac{Q_h}{Q_h - Q_c} = \frac{1}{1 - \frac{Q_c}{Q_h}} = \frac{1}{1 - \frac{T_c}{T_h}} = \frac{T_h}{T_h - T_c}\end{aligned}$$

Heat pump – Cooling mode

$$\text{COP (cooling mode)} = \frac{Q_c}{W}$$

$$W = Q_h - Q_c$$

A good refrigerator must have a high COP of 5 or 6

The greatest possible COP for a heat pump in the cooling mode is that of a heat pump whose working substance is carried through a Carnot cycle in reverse:

$$\text{COP}_C (\text{cooling mode}) = \frac{T_c}{T_h - T_c}$$

Example: If a refrigerator is to be maintained at -3°C , and the outside air is at 27°C , what minimum amount of work has to be done to remove a joule of heat from inside the refrigerator?

The minimum amount of work will be done by the most efficient refrigerator. That is a Carnot refrigerator. For a Carnot refrigerator, $|Q_{hot}|/|Q_{cold}| = T_{hot}/T_{cold}$ or $|Q_{hot}| = (T_{hot}/T_{cold})|Q_{cold}|$ and we know that $|Q_{cold}| = 1\text{ Joule}$. The work done by the refrigerator is

$$W_{frig} = |Q_{hot}| - |Q_{cold}| = \frac{T_{hot}}{T_{cold}}|Q_{cold}| - |Q_{cold}| = \left(\frac{T_{hot}}{T_{cold}} - 1\right) |Q_{cold}|. \tag{13}$$

Converting the temperatures to Kelvin ($T_{cold} = -3^{\circ}\text{C} = 270\text{ K}$ and $T_{hot} = 27^{\circ}\text{C} = 300\text{ K}$), we find $W_{frig} = 0.11\text{ J}$.

$$\begin{aligned} T_h/T_c &= Q_h/Q_c \\ &300/270 \\ Q_h &= (T_h/T_c) * Q_c \end{aligned}$$

During a cyclic process, a heat engine absorbs 500 J of heat from a hot reservoir, does work and ejects an amount of heat 300 J into the surroundings (cold reservoir). Calculate the efficiency of the heat engine?

Solution

The efficiency of heat engine is given by

$$\eta = 1 - \frac{Q_L}{Q_H}$$

$$\eta = 1 - \frac{300}{500} = 1 - \frac{3}{5}$$

$$\eta = 1 - 0.6 = 0.4$$

$$\mu = 1 - 0.6 = 0.4$$

The heat engine has 40% efficiency, implying that this heat engine converts only 40% of the input heat into work.

There are two Carnot engines A and B operating in two different temperature regions. For Engine A the temperatures of the two reservoirs are 150°C and 100°C . For engine B the temperatures of the reservoirs are 350°C and 300°C . Which engine has lesser efficiency?

Solution

The efficiency for engine A $= 1 - 373/423 = 0.11$. Engine A has 11% efficiency

The efficiency for engine B $= 1 - 573/623 = 0.08$

Engine B has only 8% efficiency.

Even though the differences between the temperature of hot and cold reservoirs in both engines is same, the efficiency is not same. The efficiency depends on the ratio of the two temperature and not on the difference in the temperature. The engine which operates in lower temperature has highest efficiency.

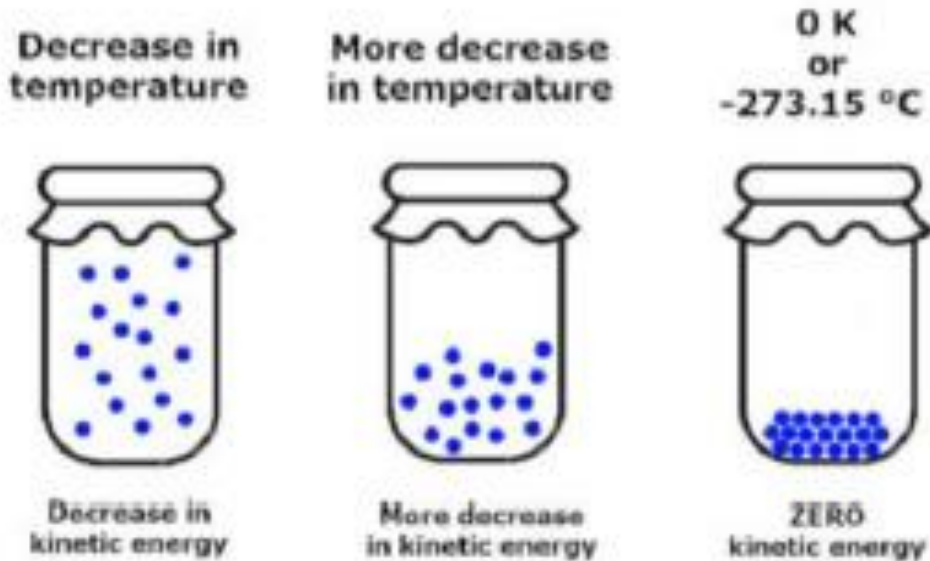
Q. A Carnot engine operates with an efficiency equals to $\frac{1}{5}$. On decreasing the temperature of the sink by 50 K, the efficiency increases to $\frac{1}{3}$. Temperatures of source and sink are respectively :

A refrigerator has COP of 3. How much work must be supplied to the refrigerator in order to remove 200 J of heat from its interior?

$$\text{COP} = \beta = \frac{Q_L}{W}$$

$$W = \frac{Q_c}{\text{COP}} = \frac{200}{3} = 66.67 \text{ J}$$

Third Law of Thermodynamics



Absolute zero temperature is the coldest possible temperature in the universe (and it is 0 K)

“The entropy of a pure crystalline substance at absolute zero is 0”

Third Law of Thermodynamics

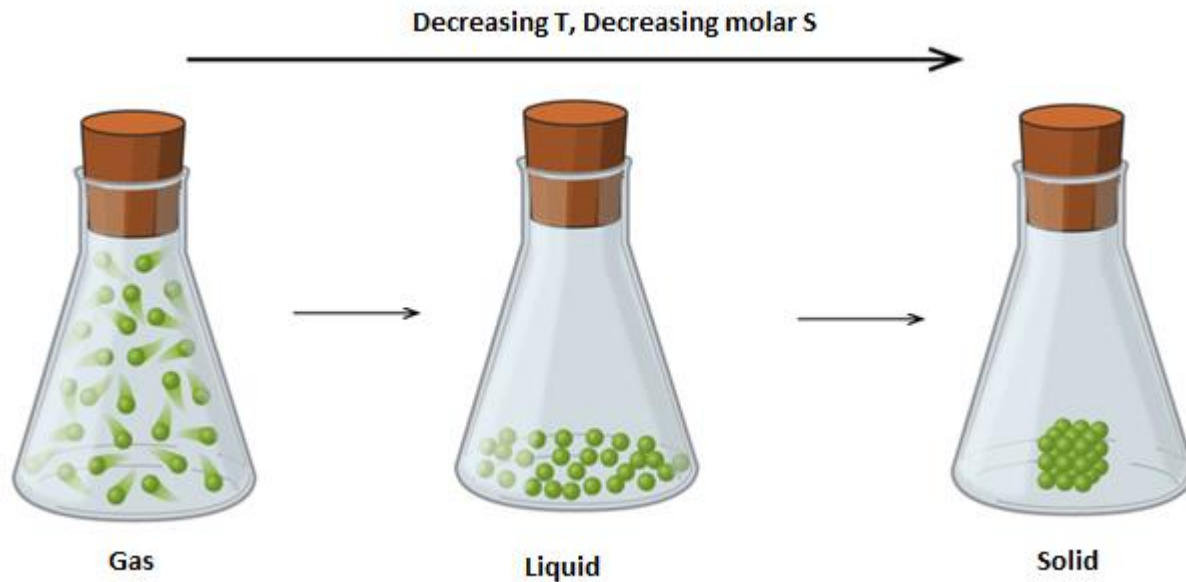
Nernst theorem: The entropy change accompanying any physical or chemical transformation approaches zero as the temperature approaches absolute zero: $\Delta S \rightarrow 0$ as $T \rightarrow 0$ (-273.15 °C, or -459.67 °F) provided all the substances involved are **perfectly crystalline**.

In practical terms, this theorem implies the **impossibility** of attaining absolute zero, since as a system approaches absolute zero, the further extraction of energy from that system becomes more and more difficult (*more work is required to extract heat from a cold body*).

Modern science has attained temperatures less than a billionth of a degree above absolute zero, **but absolute zero itself can never be reached**.

<https://www.britannica.com/biography/Walther-Nernst#ref102911>

Third Law of Thermodynamics



Disorder (randomness) of a system decreases, as we continue to extract heat from it. This results in decreasing molecular kinetic energy as we go from **gas** → **liquid** → **solid**.

Third law states that at absolute zero (0 K), the disorder in a solid must be zero.

But, practically, reaching absolute zero is not possible.

Calculating the standard entropy change for chemical changes

Illustration 3.6 *Calculating a standard reaction entropy*

To calculate the standard reaction entropy of $\text{H}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$ at 25°C, we use the data in Table 2.7 of the *Data Section* to write

$$\begin{aligned}\Delta_r S^\ominus &= S^\ominus_{\text{m}}(\text{H}_2\text{O}, \text{l}) - \{S^\ominus_{\text{m}}(\text{H}_2, \text{g}) + \tfrac{1}{2}S^\ominus_{\text{m}}(\text{O}_2, \text{g})\} \\ &= 69.9 \text{ J K}^{-1} \text{ mol}^{-1} - \{130.7 + \tfrac{1}{2}(205.0)\} \text{ J K}^{-1} \text{ mol}^{-1} \\ &= -163.4 \text{ J K}^{-1} \text{ mol}^{-1}\end{aligned}$$

The negative value is consistent with the conversion of two gases to a compact liquid.

Why **ZERO** Kelvin is not possible ?

It is not possible to achieve absolute zero temperature. According to second law of thermodynamics, heat will move from the warmer body to a colder body. So if we try to cool down the object to absolute zero, then that object will keep on receiving heat from the surrounding and so it's just a theoretical concept.

The value of entropy of a completely pure crystalline substance is zero at absolute zero temperature

Third law of thermodynamics – Microscopic Interpretation

Third law $\lim_{T \rightarrow 0} \Delta S = 0$

Quantum statistical mechanics provides the microscopic interpretation of zero entropy for a perfect crystal as well as quantitative corrections for those few errant substances exhibiting small positive entropy values at zero absolute temperature.

The statement that the lowest energy state of the crystal is nondegenerate (only a single quantum state is available to it) is easily visualized as a perfectly ordered crystal where only a single arrangement of atoms, molecules, or ions on the crystal lattice is possible.

Thus, in terms of **Boltzmann's famous equation**

$$S \equiv k_B \ln W$$

W – number of possible microstates
k_B - Boltzmann's constant

Boltzmann formula for the entropy

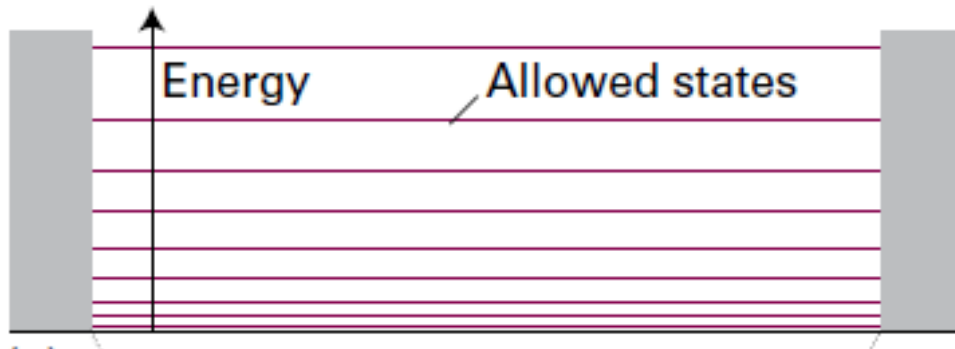
From a microscopic viewpoint, entropy is defined as $S \equiv k_B \ln W$

macroscopic properties are observable at larger scale:

pressure and temperature

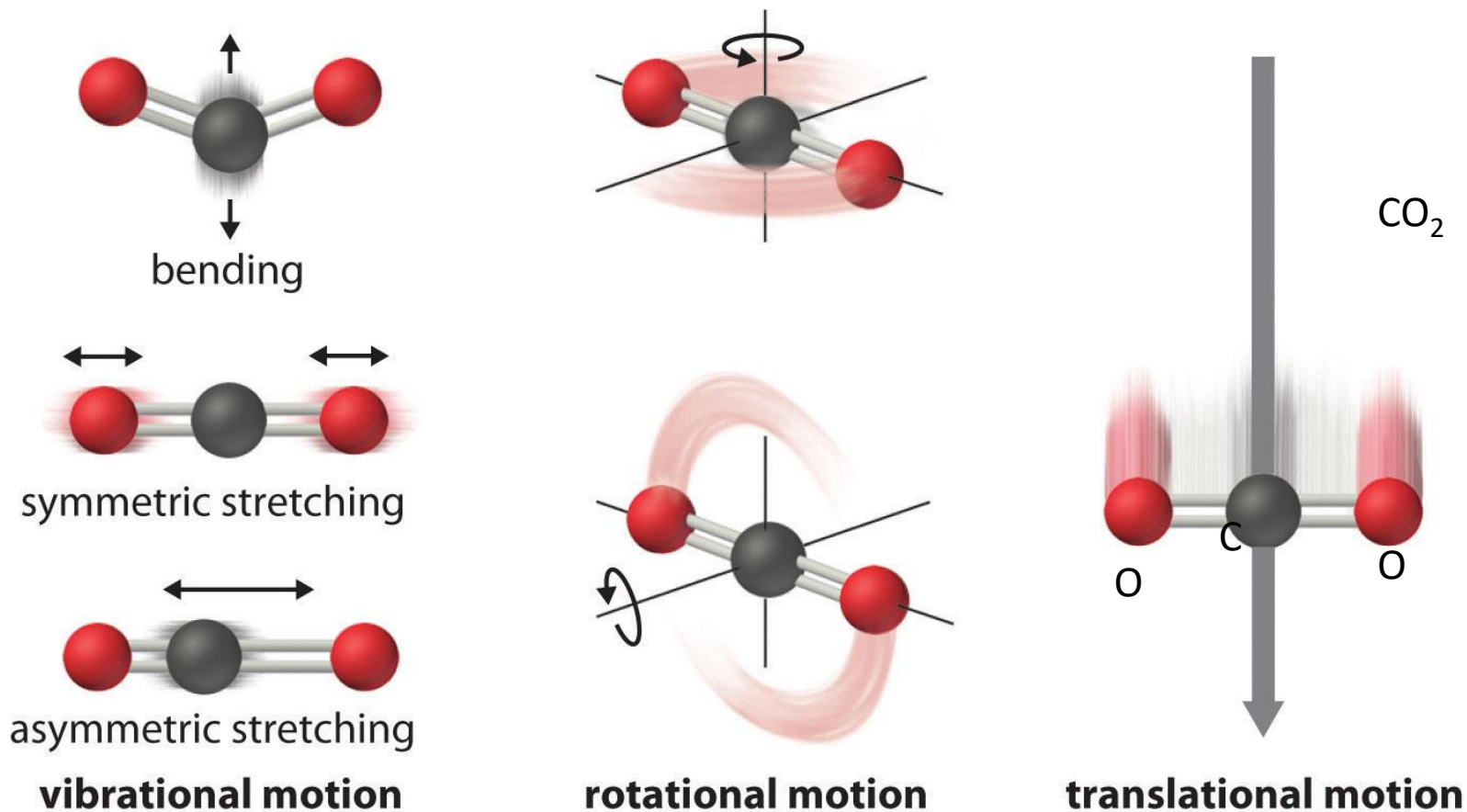
microscopic properties are observable at atomic/molecular scale:

translational, vibrational, rotational



Translational, vibrational, rotational states have quantized energy levels, (Lower energy to higher energy)

Translational, vibrational, rotational motions

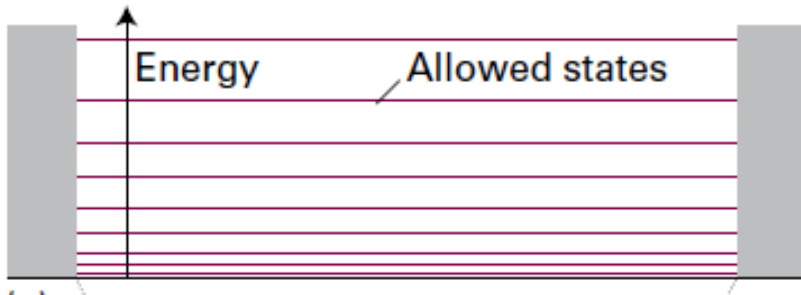


$$\lim_{T \rightarrow 0} \Delta S = 0$$

As T approaches Zero, these motions are also approaches Zero

Boltzmann formula for the entropy

From a microscopic viewpoint, entropy is defined as



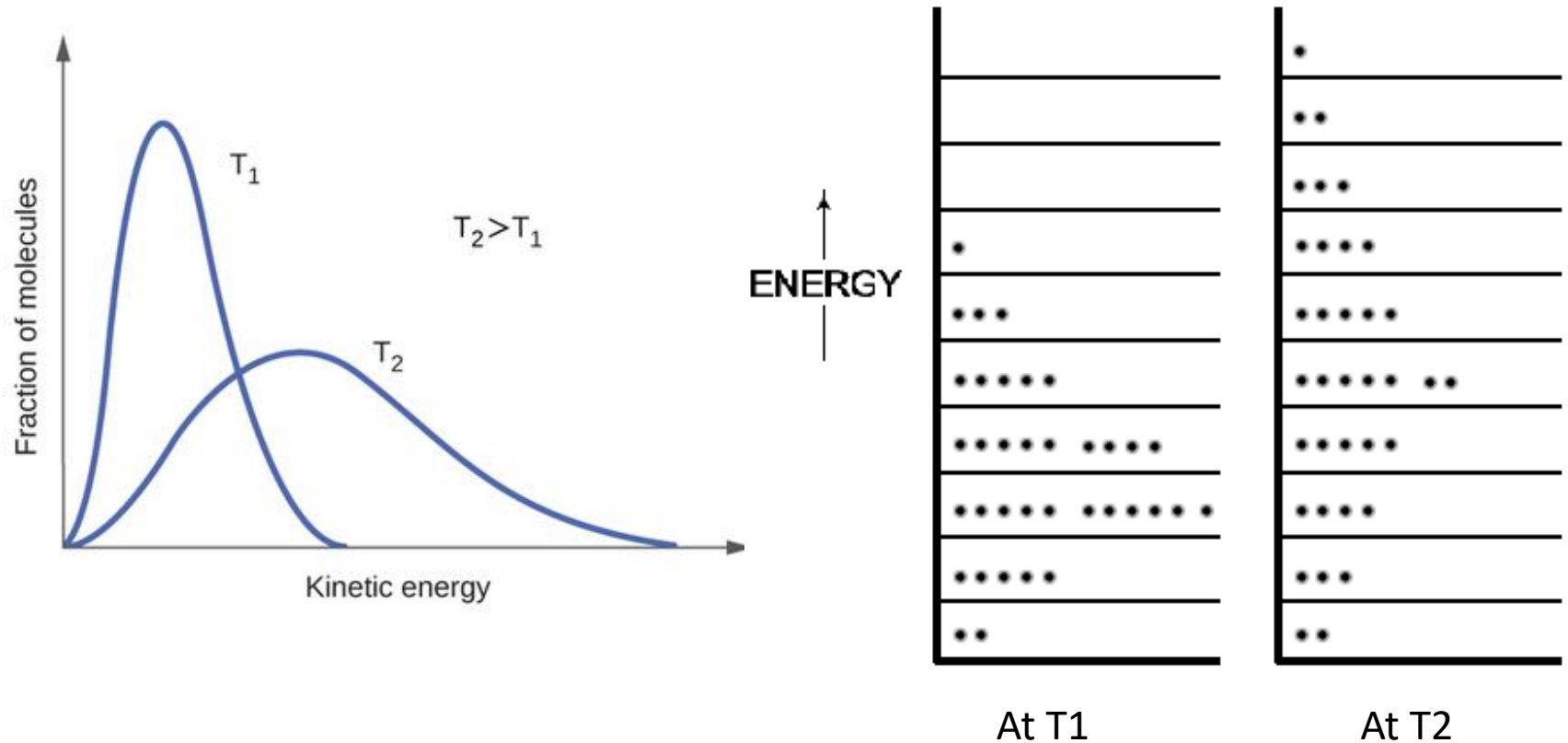
$$S \equiv k_B \ln W$$

If all the molecules are in one energy state there is only one way of achieving this distribution, so $W = 1$ and, because $\ln 1 = 0$, it follows that $S = 0$.

As the molecules spread out over the available energy states, W increases and therefore so too does the entropy.

Entropy for any substance increases with temperature

“As temperature increase the average kinetic energy of molecules increases”. Thermal energy is spread and dispersed.



Assume that there 36 molecules in a gas container.

Calculate the work done when 50 g of iron reacts with hydrochloric acid to produce $\text{FeCl}_2(\text{aq})$ and hydrogen in (a) a closed vessel of fixed volume, (b) an open beaker at 25°C .

The solution In (a) the volume cannot change, so no expansion work is done and $w = 0$. In (b) the gas drives back the atmosphere and therefore $w = -p_{\text{ex}}\Delta V$. The initial volume can be neglected because the final volume (after the production of gas) is so much larger and $\Delta V = V_{\text{f}} - V_{\text{i}} \approx V_{\text{f}} = nRT/p_{\text{ex}}$, where n is the amount of H_2 produced. Therefore,

$$w = -p_{\text{ex}}\Delta V \approx -p_{\text{ex}} \times \frac{nRT}{p_{\text{ex}}} = -nRT$$

Because the reaction is $\text{Fe}(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{FeCl}_2(\text{aq}) + \text{H}_2(\text{g})$, 1 mol H_2 is generated when 1 mol Fe is consumed, and n can be taken as the amount of Fe atoms that react. Because the molar mass of Fe is 55.85 g mol^{-1} , it follows that

$$\begin{aligned} w &= -\frac{50\text{ g}}{55.85\text{ g mol}^{-1}} \times (8.3145\text{ J K}^{-1}\text{ mol}^{-1}) \times (298\text{ K}) \\ &\approx -2.2\text{ kJ} \end{aligned}$$

When a sample of 1.00 mol Ar, regarded here as a perfect gas, undergoes an isothermal reversible expansion at 20.0 °C from 10.0 dm³ to 30.0 dm³ the work done is

$$\begin{aligned}w &= -(1.00 \text{ mol}) \times (8.3145 \text{ J K}^{-1} \text{ mol}^{-1}) \times (293.2 \text{ K}) \ln \frac{30.0 \text{ dm}^3}{10.0 \text{ dm}^3} \\&= -2.68 \text{ kJ}\end{aligned}$$

When a sample of argon (for which $\gamma = \frac{5}{3}$) at 100 kPa expands reversibly and adiabatically to twice its initial volume the final pressure will be

$$p_f = \left(\frac{V_i}{V_f} \right)^\gamma p_i = \left(\frac{1}{2} \right)^{5/3} \times (100 \text{ kPa}) = 31 \text{ kPa}$$

To calculate the entropy change in the surroundings when 1.00 mol $\text{H}_2\text{O}(\text{l})$ is formed from its elements under standard conditions at 298 K, use $\Delta_{\text{f}}H^{\ominus} = -286 \text{ kJ mol}^{-1}$ from Table 2C.4. The energy released as heat from the system is supplied to the surroundings, so $q_{\text{sur}} = +286 \text{ kJ}$. Therefore,

$$\Delta S_{\text{sur}} = \frac{2.86 \times 10^5 \text{ J}}{298 \text{ K}} = +960 \text{ J K}^{-1}$$

This strongly exothermic reaction results in an increase in the entropy of the surroundings as energy is released as heat into them.

The Carnot cycle can be regarded as a representation of the changes taking place in a heat engine in which part of the energy extracted as heat from the hot reservoir is converted into work. Consider an engine running in accord with the Carnot cycle, and in which 100 J of energy is withdrawn from the hot source ($q_h = -100 \text{ J}$) at 500 K. Some of this energy is used to do work and the remainder is deposited in the cold sink at 300 K. According to eqn 3A.6, the heat deposited is

$$q_c = -q_h \times \frac{T_c}{T_h} = -(-100 \text{ J}) \times \frac{300 \text{ K}}{500 \text{ K}} = +60 \text{ J}$$

This value implies that 40 J was used to do work.

Example: Problem 18.27. The temperature at the surface of the Sun is approximately 5700 K, and the temperature at the surface of the Earth is approximately 290 K. What entropy change occurs when 1000 J of energy is transferred by radiation from the Sun to the Earth?

1000 J of energy is such a small amount of energy that the temperature of the Sun and Earth don't change. We use $\Delta S = Q/T$. The Sun loses heat, so $Q_{sun} < 0$, which means that its entropy decreases. $\Delta S_{Sun} = Q_{Sun}/T_{Sun} = -1000 \text{ J}/5700 \text{ K}$. The Earth gains heat, so $Q_{Earth} > 0$, which means that the Earth's entropy increases. $\Delta S_{Earth} = Q_{Earth}/T_{Earth} = 1000 \text{ J}/290 \text{ K}$. The net change in entropy is the sum of the changes (which winds up being a difference since the Sun loses entropy and the Earth gains entropy):

$$\Delta S = \Delta S_{Sun} + \Delta S_{Earth} = \frac{Q_{Sun}}{T_{Sun}} + \frac{Q_{Earth}}{T_{Earth}} = \frac{-1000 \text{ J}}{5700 \text{ K}} + \frac{1000 \text{ J}}{290 \text{ K}} = 3.27 \text{ J/K} \quad (37)$$

Example: Problem 18.28. A 1.00 kg iron horseshoe is taken from a forge at 900°C and dropped into 4.00 kg of water at 10.0°C . Assuming that no energy is lost by heat to the surroundings, determine the total entropy change of the horseshoe–water system.

We use

$$\Delta S = \Delta S_{\text{water}} + \Delta S_{\text{iron}} = \int \frac{dQ_{\text{water}}}{T} + \int \frac{dQ_{\text{iron}}}{T} \quad (38)$$

We leave the temperature under the integral sign because the temperature changes as the iron cools and the water heats up. We can express dQ as follows. Recall that $Q = mC\Delta T$. So $dQ_{\text{water}} = m_{\text{water}}C_{\text{water}}dT$ and $dQ_{\text{iron}} = m_{\text{iron}}C_{\text{iron}}dT$.

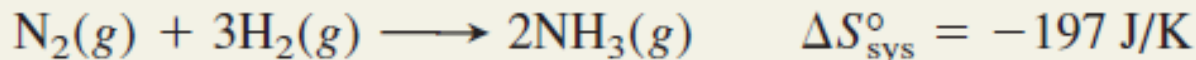
$$\begin{aligned} \Delta S &= \Delta S_{\text{water}} + \Delta S_{\text{iron}} \\ &= \int \frac{dQ_{\text{water}}}{T} + \int \frac{dQ_{\text{iron}}}{T} \\ &= \int_{T_{i,\text{water}}}^{T_f} \frac{m_{\text{water}}C_{\text{water}}dT}{T} + \int_{T_{i,\text{iron}}}^{T_f} \frac{m_{\text{iron}}C_{\text{iron}}dT}{T} \\ &= m_{\text{water}}C_{\text{water}} \ln \left(\frac{T_f}{T_{i,\text{water}}} \right) + m_{\text{iron}}C_{\text{iron}} \ln \left(\frac{T_f}{T_{i,\text{iron}}} \right) \end{aligned} \quad (39)$$

where T_f is the final temperature. Both the iron and the water reach the same final temperature. To calculate the final temperature, we use $Q_{\text{cold}} = -Q_{\text{hot}}$. This leads to

$$\begin{aligned} m_{\text{water}}C_{\text{water}}\Delta T_{\text{water}} &= -m_{\text{iron}}C_{\text{iron}}\Delta T_{\text{iron}} \\ m_{\text{water}}C_{\text{water}}(T_f - 10.0^\circ\text{C}) &= -m_{\text{iron}}C_{\text{iron}}(T_f - 900^\circ\text{C}) \end{aligned} \quad (40)$$

Using $C_{\text{water}} = 4186\text{ J/kg}\cdot^\circ\text{C}$ and $C_{\text{iron}} = 448\text{ J/kg}\cdot^\circ\text{C}$, we find $T_f = 33.2^\circ\text{C} = 306.2\text{ K}$. Plugging this into eq. (39) yields $\Delta S = 718\text{ J/K}$.

PROBLEM At 298 K, the formation of ammonia has a negative $\Delta S_{\text{sys}}^{\circ}$:



Calculate ΔS_{univ} , and state whether the reaction occurs spontaneously at this temperature.

PLAN For the reaction to occur spontaneously, $\Delta S_{\text{univ}} > 0$, and so ΔS_{surr} must be greater than $+197 \text{ J/K}$. To find ΔS_{surr} , we need $\Delta H_{\text{sys}}^{\circ}$, which is the same as $\Delta H_{\text{rxn}}^{\circ}$. We use $\Delta H_{\text{f}}^{\circ}$ values from Appendix B to find $\Delta H_{\text{rxn}}^{\circ}$. Then, we use $\Delta H_{\text{rxn}}^{\circ}$ and the given T (298 K) to find ΔS_{surr} . To find ΔS_{univ} , we add the calculated ΔS_{surr} to the given $\Delta S_{\text{sys}}^{\circ}$ (-197 J/K).

SOLUTION Calculating $\Delta H_{\text{sys}}^{\circ}$:

$$\begin{aligned}\Delta H_{\text{sys}}^{\circ} &= \Delta H_{\text{rxn}}^{\circ} \\ &= [(2 \text{ mol NH}_3)(-45.9 \text{ kJ/mol})] - [(3 \text{ mol H}_2)(0 \text{ kJ/mol}) + (1 \text{ mol N}_2)(0 \text{ kJ/mol})] \\ &= -91.8 \text{ kJ}\end{aligned}$$

Calculating ΔS_{surr} :

$$\Delta S_{\text{surr}} = -\frac{\Delta H_{\text{sys}}^{\circ}}{T} = -\frac{-91.8 \text{ kJ} \times \frac{1000 \text{ J}}{1 \text{ kJ}}}{298 \text{ K}} = 308 \text{ J/K}$$

Determining ΔS_{univ} :

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}}^{\circ} + \Delta S_{\text{surr}} = -197 \text{ J/K} + 308 \text{ J/K} = 111 \text{ J/K}$$

$\Delta S_{\text{univ}} > 0$, so the reaction occurs spontaneously at 298 K (see figure in margin).

0th Law : Defines temperature (T)

1st Law : Defines energy (E)

2nd Law : Defines entropy (S)

3rd Law : Gives value for entropy

Summary

- Isothermal process: Temperature is constant ($dT = 0$)
- Adiabatic process: no heat is exchanged ($q = 0$)
- Zeroth law of thermodynamics
- First law of thermodynamics
 - Work done by gas at constant pressure: $w = -p dV$
 - Different types of work done

$$V T^\gamma = \text{constant}$$

$$p_f V_f^\gamma = p_i V_i^\gamma$$

$$\Delta U = Q - W$$

$$\Delta H = \Delta U + p dV$$

$$dU = C_v dT$$

$$dH = C_p dT$$

- Second Law of Thermodynamics

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$$

$$\Delta G = \Delta H - T \Delta S$$

$$\Delta S = \frac{Q}{T},$$

- 3rd Law of Thermodynamics

$$\lim_{T \rightarrow 0} \Delta S = 0$$

Summary

- Heat engine changes heat into useful work; needs temperature difference.

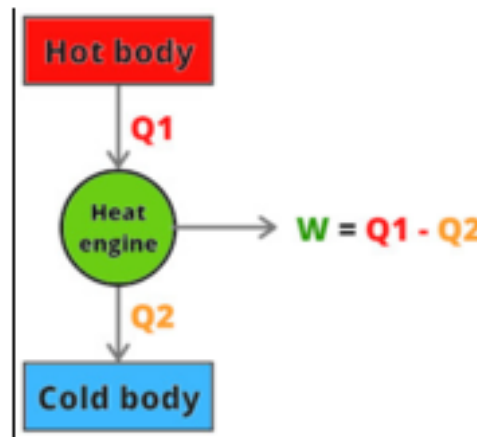
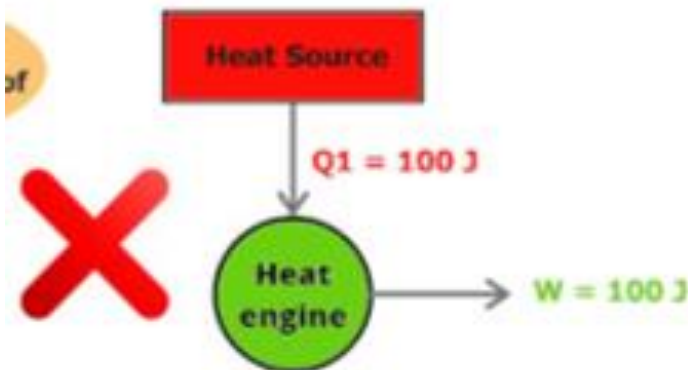
- Efficiency of a heat engine:

$$e = \frac{W}{Q_H}.$$

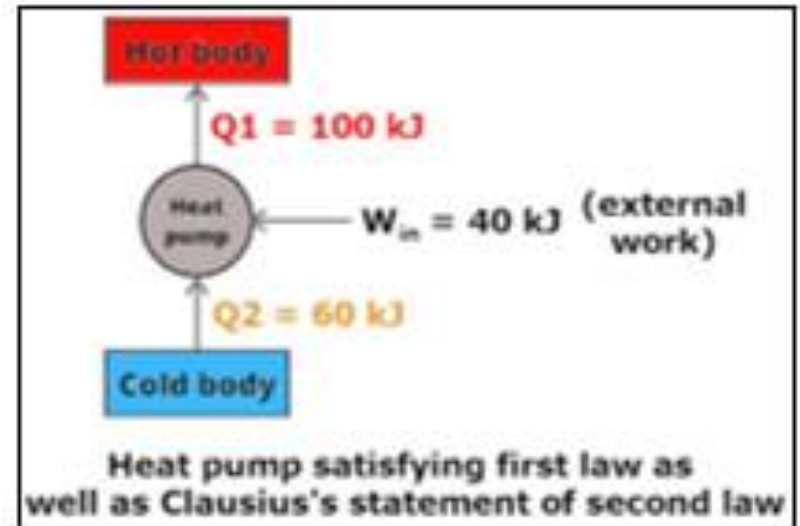
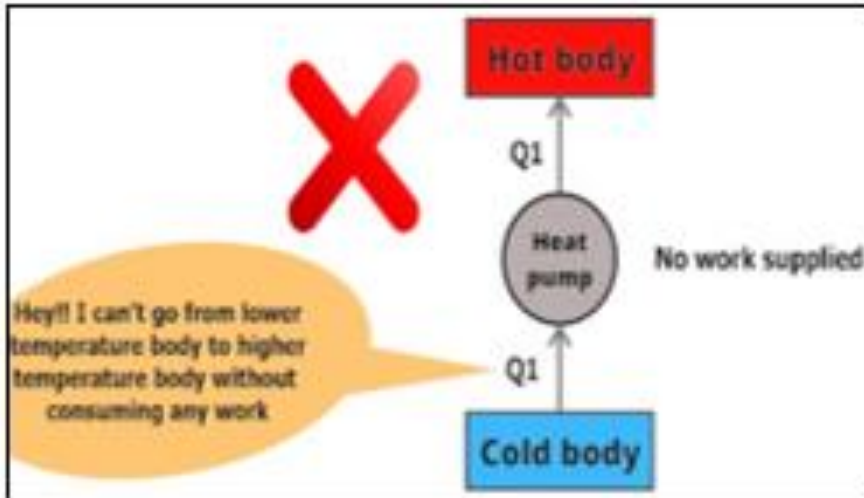
$$e = \frac{W}{Q_H} = \frac{Q_H - Q_L}{Q_H}$$

$$e = 1 - \frac{Q_L}{Q_H}.$$

$$= 1 - \frac{T_L}{T_H}$$



Summary



Summary

- Refrigerators and air conditioners do work to extract heat from a cooler region and send it to a warmer region:
- A heat pump is similar:

$$\text{COP} = \frac{Q_L}{W} \quad \left[\begin{array}{l} \text{refrigerator and} \\ \text{air conditioner} \end{array} \right] \quad (15-6a)$$

$$\text{COP} = \frac{Q_H}{W} \quad [\text{heat pump}] \quad (15-7)$$

Third Law of thermodynamics

- At $T = 0$, all energy of thermal motion has been quenched, and in a perfect crystal all the atoms or ions are in a regular, uniform array .

then there is only one way of arranging the molecules when they are all in the ground state, which is the case at $T = 0$. Thus, at $T = 0$, $W = 1$ and from $S = k \ln W$ it follows that $S = 0$.

The entropy of all perfect crystalline substances is zero at $T = 0$

(Third Law of thermodynamics)

Further Reading

Entropy and randomness

Entropy (S) has units of $\text{J K}^{-1} \text{mol}^{-1}$, same unit as gas constant R

On a macroscopic scale, we "quantify" entropy by measuring heat flow. The change in entropy is defined as the "reversible" heat divided by the kelvin temperature (K).

$$\Delta S = q_{\text{rev}}/T$$

Finally, entropy change is related to *the "amount" of thermal energy (q)* in a sample (system).

As the *energy "leaves" ($q < 0$)*, then the entropy of the system will decrease.

As the *energy "comes in" ($q > 0$)*, the entropy of the system will increase.

"More the energy content in a system ($q > 0$), more will be disorder"

■ Atomic size

Standard molar entropy increases down the group.

It is due to the increase of atomic size or molecular size.

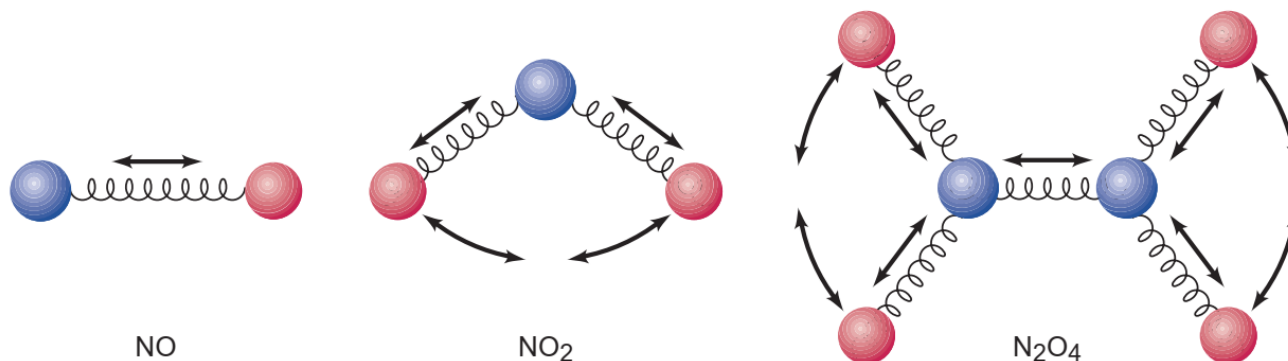
	Li	Na	K	Rb	Cs
Atomic radius (pm):	152	186	227	248	265
Molar mass (g/mol):	6.941	22.99	39.10	85.47	132.9
$S^\circ(s)$:	29.1	51.4	64.7	69.5	85.2

	HF	HCl	HBr	HI
Molar mass (g/mol):	20.01	36.46	80.91	127.9
$S^\circ(g)$:	173.7	186.8	198.6	206.3

Molecular complexity

As the molecular complexity increases, standard molar entropy increases. It is due to the efficient dispersal of energy in molecular vibrations.

	NO	NO ₂	N ₂ O ₄
$S^\circ(s):$			
$S^\circ(g):$	211	240	304

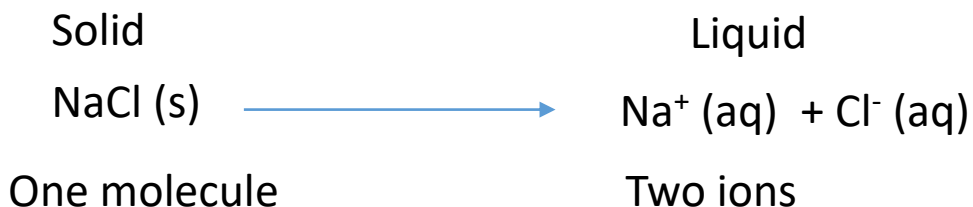


	CH ₄ (g)	C ₂ H ₆ (g)	C ₃ H ₈ (g)	C ₄ H ₁₀ (g)	C ₅ H ₁₀ (g)	C ₅ H ₁₀ (cyclo, g)	C ₂ H ₅ OH(l)
$S^\circ:$	186	230	270	310	348	293	161

■ Dissolving a solid or liquid

The entropy of a dissolved molecular solid or liquid is greater than the entropy of the pure solute, some exceptions may be found (e.g. AlCl_3)

	NaCl	AlCl_3	CH_3OH
$S^\circ(s \text{ or } l):$	72.1(s)	167(s)	127(l)
$S^\circ(aq):$	115.1	-148	132



■ Dissolving a gas

The particles in a gas already have so much freedom of motion—and, thus, such highly dispersed energy—that they lose some when they dissolve in a liquid or solid. Therefore, the entropy of a solution of a gas in a liquid or in a solid is always *less* than the entropy of the gas itself.



For dissolving a gas in liquid

$$\Delta S = S_{\text{final}} - S_{\text{initial}}$$

$$\Delta S = S_{\text{liq}} - S_{\text{gas}}$$

$$S_{\text{liq}} \ll S_{\text{gas}}$$

$$\text{So } \Delta S < 0$$

Predicting the sign of the entropy change of system

Predict the sign of the entropy change for the following processes. Indicate the reason for each of your predictions.

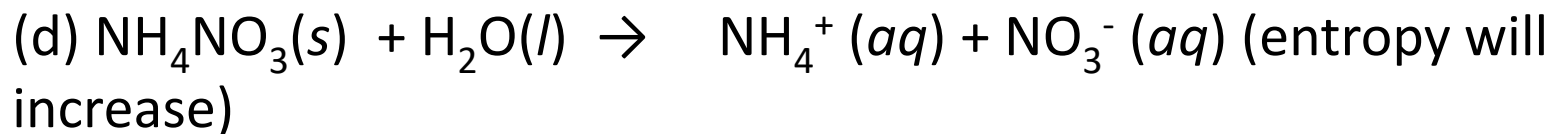
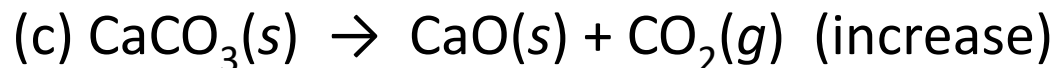
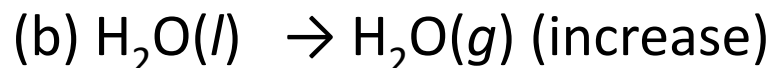
1. One mole liquid water at room temperature $\rightarrow$ one mole liquid water at 50 °C
2. $\text{Ag}^+(aq) + \text{Cl}^-(aq) \rightarrow \text{AgCl}(s)$
3. $\text{C}_6\text{H}_6(l) + 15/2 \text{O}_2(g) \rightarrow 6\text{CO}_2(g) + 3\text{H}_2\text{O}(l)$
4. $\text{NH}_3(s) \rightarrow \text{NH}_3(l)$

$$\Delta S_{\text{sys}} = S_{\text{final}} - S_{\text{initial}}$$

Solution

1. Positive, as the temperature increases molecular motions raises.
2. Negative, reduction in the number of ions (particles) in solution, decreased dispersal of matter
3. Negative, net decrease in the amount of gaseous species
4. Positive, phase transition from solid to liquid, net increase in dispersal of matter

Which of the following processes will lead to an increase in the entropy of the system?



SAMPLE PROBLEM 20.1 Predicting Relative Entropy Values

PROBLEM Choose the member with the higher entropy in each of the following pairs, and justify your choice [assume constant temperature, except in part (e)]:

- (a) 1 mol of $\text{SO}_2(g)$ or 1 mol of $\text{SO}_3(g)$
- (b) 1 mol of $\text{CO}_2(s)$ or 1 mol of $\text{CO}_2(g)$
- (c) 3 mol of $\text{O}_2(g)$ or 2 mol of $\text{O}_3(g)$
- (d) 1 mol of $\text{KBr}(s)$ or 1 mol of $\text{KBr}(aq)$
- (e) Seawater at 2°C or at 23°C
- (f) 1 mol of $\text{CF}_4(g)$ or 1 mol of $\text{CCl}_4(g)$

SOLUTION (a) 1 mol of $\text{SO}_3(g)$. For equal numbers of moles of substances with the same types of atoms in the same physical state, the more atoms in the molecule, the more types of motion available, and thus the higher the entropy.

(b) 1 mol of $\text{CO}_2(g)$. For a given substance, entropy increases in the sequence $s < l < g$.

(c) 3 mol of $\text{O}_2(g)$. The two samples contain the same number of oxygen atoms but different numbers of molecules. Despite the greater complexity of O_3 , the greater number of molecules dominates in this case because there are many more microstates possible for three moles of particles than for two moles.

(d) 1 mol of $\text{KBr}(aq)$. The two samples have the same number of ions, but their motion is more limited and their energy less dispersed in the solid than in the solution.

(e) Seawater at 23°C . Entropy increases with rising temperature.

(f) 1 mol of $\text{CCl}_4(g)$. For similar compounds, entropy increases with molar mass.

FOLLOW-UP PROBLEM 20.1 For 1 mol of substance at a given temperature, select the member with the higher entropy in each of the following pairs, and give the reason for your choice:

(a) $\text{PCl}_3(g)$ or $\text{PCl}_5(g)$ (b) $\text{CaF}_2(s)$ or $\text{BaCl}_2(s)$ (c) $\text{Br}_2(g)$ or $\text{Br}_2(l)$

Factors affecting entropy

1. Temperature changes. For a given substance, S° increases as the temperature rises.
2. Physical states and phase changes. entropy increases for a substance as it changes from a solid to a liquid to a gas
3. Dissolving a solid or liquid. The entropy of a dissolved molecular solid or liquid is greater than the entropy of the pure solute.
4. Dissolving a gas. The entropy of a solution of a gas in a liquid or in a solid is always less than the entropy of the gas itself.
5. Molecular complexity. For compounds, entropy increases with chemical complexity, that is, with the number of atoms in a formula unit or molecule of the compound.
6. Atomic size. Within a periodic group, molar entropy increases down the group:

Free energy change-Non-expansion work

Change in Entropy and Spontaneity

$$\Delta S_{\text{univ}} = +\text{Ve (Spontaneous)}$$

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$$

However, the change in entropy of surrounding is difficult to measure experimentally.

Therefore we need to introduce a quantity/function called, Gibbs free energy or free energy (G) that combines systems enthalpy and entropy:

$$G = H - TS$$

At constant T, the change in free energy is

$$\Delta G = \Delta H - T\Delta S$$

■ Derivation of $\Delta G = \Delta H_{\text{sys}} - T\Delta S_{\text{sys}}$

It begins with our original hypothesis.

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$$

At constant pressure,

$$\Delta S_{\text{surr}} = -\frac{\Delta H_{\text{sys}}}{T}$$

Substituting for ΔS_{surr} gives a relationship that lets us focus solely on the system:

$$-T \Delta S_{\text{univ}} = \left(\Delta S_{\text{sys}} - \frac{\Delta H_{\text{sys}}}{T} \right) \quad (-T)$$

Multiplying both sides by $-T$ gives

$$-T\Delta S_{\text{univ}} = \Delta H_{\text{sys}} - T\Delta S_{\text{sys}}$$

Combining this equation with the previous one shows that

$$-T\Delta S_{\text{univ}} = \Delta H_{\text{sys}} - T\Delta S_{\text{sys}} = \Delta G_{\text{sys}}$$

*The **sign** of ΔG tells if a reaction is spontaneous.* The second law dictates

- $\Delta S_{\text{univ}} > 0$ for a spontaneous process
- $\Delta S_{\text{univ}} < 0$ for a nonspontaneous process
- $\Delta S_{\text{univ}} = 0$ for a process at equilibrium

Of course, absolute temperature is always positive, so

$$T\Delta S_{\text{univ}} > 0 \quad \text{or} \quad -T\Delta S_{\text{univ}} < 0 \text{ for a spontaneous process}$$

Because $\Delta G = -T\Delta S_{\text{univ}}$, we know that

- $\Delta G < 0$ for a spontaneous process
- $\Delta G > 0$ for a nonspontaneous process
- $\Delta G = 0$ for a process at equilibrium

Relative Importance of ΔH and ΔS in Spontaneity

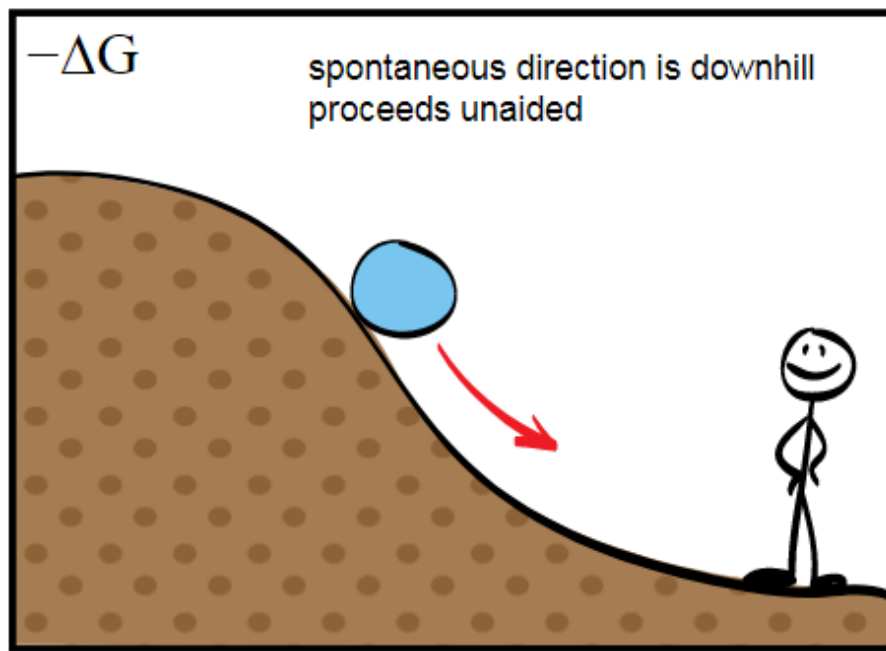
$$\Delta G_{\text{sys}} = \Delta H_{\text{sys}} - T\Delta S_{\text{sys}}$$

The four possible combinations for the signs of ΔH and ΔS

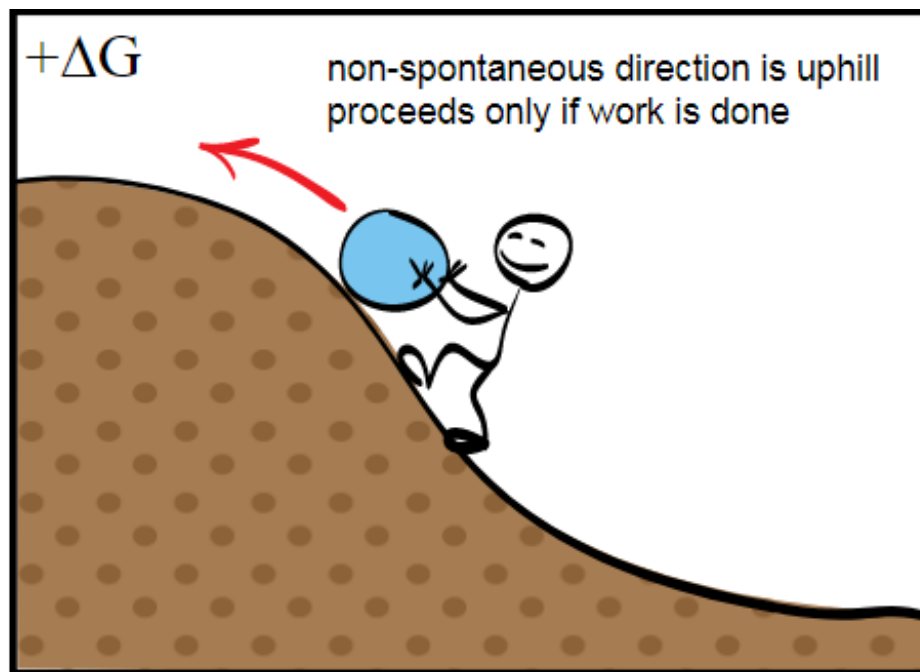
Sign of ΔH	Sign of ΔS	Implications for Spontaneity
–	+	Spontaneous at all temperatures
+	–	Never spontaneous
–	–	Spontaneous only at low temperatures
+	+	Spontaneous only at high temperatures

Enthalpy and entropy compensation effect

If something (reaction or process) lowers the free energy, we say such a process is "spontaneous." That simply means it will tend to happen - much like a ball *tends* to roll downhill.



The reverse of this situation is also evident - if a process is "not spontaneous," then it will *tend* to not happen.



This is not to say that non-spontaneous processes/reactions *never* happen, it is just that if they are to happen, some external amount of work must be applied to make the process happen - much like physically taking the ball up the hill to a higher position

ΔG and non-expansion work

Apart from predicting spontaneity, a second **feature of the Gibbs energy** is that *the value of ΔG for a process gives the **maximum non-expansion work** that can be extracted from the process at constant temperature and pressure.*

By **non-expansion work**, w , we mean any work other than that arising from the expansion of the system.

At constant temperature and pressure:

$$\Delta G = w_{\max}$$

Example;

Electrochemical cells, $\Delta G = -nFE^\circ = w_{\max}$

Summary of Free energy and Entropy

The *only* criterion of spontaneous change is the total entropy of the system and its surroundings must be positive;

The **Gibbs energy just provides a way to express the total entropy change** in terms of the properties of the **system alone**, and is **valid only for processes that occur at constant temperature and pressure**.

- **Reaction enthalpy** : The enthalpy change accompanying a reaction

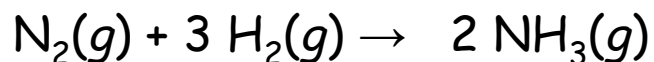
$$\Delta H^{\circ} = \Sigma H_f^{\circ}(\text{products}) - \Sigma H_f^{\circ}(\text{reactants})$$

- **Reaction entropy** : The entropy change accompanying a reaction

$$\Delta S^{\circ} = \Sigma S^{\circ}(\text{products}) - \Sigma S^{\circ}(\text{reactants})$$

Problem 1

Calculate ΔH_f° and ΔS_f° for the following reaction and decide in which direction each of these factors will drive the reaction.



Compound	$H_f^\circ(\text{kJ/mol})$	$S^\circ(\text{J/mol-K})$
$\text{N}_2(g)$	0	191.61
$\text{H}_2(g)$	0	130.68
$\text{NH}_3(g)$	-46.11	192.45

Solution 1

$$\Delta H^\circ = \Sigma H_f^\circ(\text{products}) - \Sigma H_f^\circ(\text{reactants})$$

$$= [2 \text{ mol NH}_3 \times 46.11 \text{ kJ/mol}] - [1 \text{ mol N}_2 \times 0 \text{ kJ/mol} + 3 \text{ mol H}_2 \times 0 \text{ kJ/mol}]$$

$$= -92.22 \text{ kJ}$$

$$\Delta S^\circ = \Sigma S^\circ(\text{products}) - \Sigma S^\circ(\text{reactants})$$

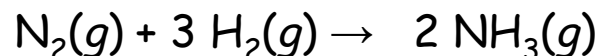
$$= [2 \text{ mol NH}_3 \times 192.45 \text{ J/mol-K}] - [1 \text{ mol N}_2 \times 191.61 \text{ J/mol-K} + 3 \text{ mol H}_2 \times 130.68 \text{ J/mol-K}]$$

$$= -198.75 \text{ J/K}$$

Since ΔH° and ΔS° are negative, the spontaneity in right direction must be temperature dependent. This can be analyzed by calculating ΔG° at different temperatures.

Problem 2

Predict direction of this reaction at 25 °C and 500 °C.



Solutions

At 25°C

$$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ$$

$$= -92,220 \text{ J} - (298.15 \text{ K} \times -198.75 \text{ J/K})$$

$$= -92,220 \text{ J} + 59,260 \text{ J}$$

$$= -32,960 \text{ J}$$

At 500 °C

$$\Delta G^\circ_{773} = \Delta H^\circ_{298} - T \Delta S^\circ_{298}$$

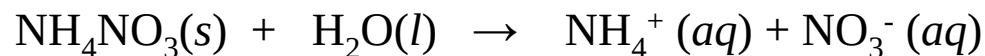
$$= 92,220 \text{ J} - (773 \text{ K} \times -198.75 \text{ J/K})$$

$$= 92,220 \text{ J} - (-153,600 \text{ J})$$

$$= 61,380 \text{ J}$$

Problem 3

Predict spontaneity of reaction by calculating ΔS° and ΔH°



Compound	$H_f^\circ(\text{kJ/mol})$	$S^\circ(\text{J/mol-K})$
$\text{NH}_4\text{NO}_3(s)$	-365.56	151.08
$\text{NH}_4^+(aq)$	-132.51	113.4
$\text{NO}_3^-(aq)$	-205.0	146.4

Solutions

$$\Delta H^\circ = \Sigma H_f^\circ(\text{products}) - \Sigma H_f^\circ(\text{reactants})$$

$$\begin{aligned} &= [1 \text{ mol NH}_4 \times 132.51 \text{ kJ/mol} + 1 \text{ mol NO}_3^- \times -205.0 \text{ kJ/mol}] - [1 \text{ mol NH}_4\text{NO}_3 \times -365.56 \text{ kJ/mol}] \\ &= 28.05 \text{ kJ} \end{aligned}$$

$$\Delta S^\circ = \Sigma S^\circ(\text{products}) - \Sigma S^\circ(\text{reactants})$$

$$\begin{aligned} &= [1 \text{ mol NH}_4 \times 113.4 \text{ J/mol-K} + 1 \text{ mol NO}_3^- \times 146.4 \text{ J/mol-K}] - [1 \text{ mol NH}_4\text{NO}_3 \times 151.08 \text{ J/mol-K}] \\ &= 108.7 \text{ J/K} \end{aligned}$$

Based on this we can't judge whether this reaction is spontaneous.

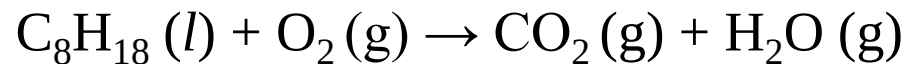
Therefore we need to calculate ΔG°

Solution 3 (Continued)

$$\begin{aligned}\Delta G^{\circ} &= \Delta H^{\circ} - T \Delta S^{\circ} \\ &= 28,050 \text{ J} - (298.15 \text{ K} \times 108.7 \text{ J/K}) \\ &= 28,050 \text{ J} - 32,410 \text{ J} \\ &= -4360 \text{ J}\end{aligned}$$

Problem 4

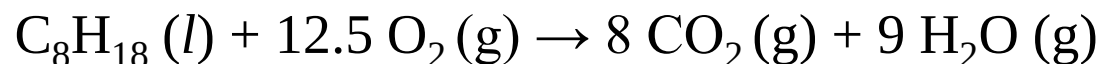
Calculate ΔS° for the combustion reaction of liquid isooctane



Compound	$S^\circ(\text{J/mol-K})$
C_8H_{18}	329.3
O_2	205.14
CO_2	213.74
H_2O	188.83

Solutions

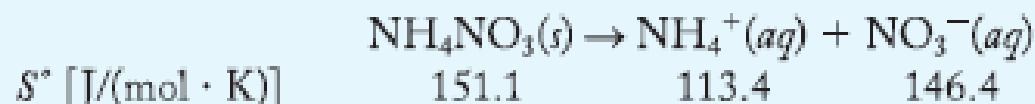
Balance the chemical reaction:



$$\begin{aligned}\Delta S_{\text{rxn}}^{\circ} &= \sum m S^{\circ}(\text{products}) - \sum n S^{\circ}(\text{reactants}) \\&= [8 S^{\circ}(\text{CO}_2) + 9 S^{\circ}(\text{H}_2\text{O})] - [S^{\circ}(\text{C}_8\text{H}_{18}) + \frac{25}{2} S^{\circ}(\text{O}_2)] \\&= \{[8 \text{ mol CO}_2 \times 213.8 \text{ J}/(\text{mol} \cdot \text{K})] + [9 \text{ mol H}_2\text{O} \times 188.8 \text{ J}/(\text{mol} \cdot \text{K})]\} \\&\quad - \left\{ [1 \text{ mol C}_8\text{H}_{18} \times 329.3 \text{ J}/(\text{mol} \cdot \text{K})] + \left[\frac{25}{2} \text{ mol O}_2 \times 205.2 \text{ J}/(\text{mol} \cdot \text{K}) \right] \right\} \\&= 515.3 \text{ J/K}\end{aligned}$$

Problem 5

What is ΔS° when ammonium nitrate dissolves (Figure 17.10, $\Delta H^\circ > 0$) under standard conditions, given the following standard molar entropy values:



Solve

$$\begin{aligned}\Delta S^\circ &= \sum n_{\text{products}} S_{\text{products}}^\circ - \sum n_{\text{reactants}} S_{\text{reactants}}^\circ \\ &= \left[1 \text{ mol} \times \left(\frac{113.4 \text{ J}}{\text{mol} \cdot \text{K}} \right) + 1 \text{ mol} \times \left(\frac{146.4 \text{ J}}{\text{mol} \cdot \text{K}} \right) \right] - 1 \text{ mol} \times \left(\frac{151.1 \text{ J}}{\text{mol} \cdot \text{K}} \right) \\ &= 108.7 \text{ J/K}\end{aligned}$$

Problem 6

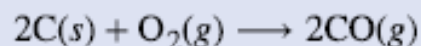
For the reaction, $2 \text{H}_2 (\text{g}) + \text{O}_2 (\text{g}) \rightarrow 2 \text{H}_2\text{O} (\text{l})$

- (a) Predict the sign of entropy change for the reaction
- (b) Is the reaction spontaneous at 298 K

Compound	$S^\circ(\text{J/mol-K})$	$H_f^\circ(\text{kJ/mol})$
H_2	130.68	0
O_2	205.14	0
$\text{H}_2\text{O} (\text{l})$	69.91	- 285.8
$\text{H}_2\text{O} (\text{g})$	188.83	-241.82

Predicting the Temperature Dependence of Spontaneity

The incomplete combustion of carbon is described by the following equation:



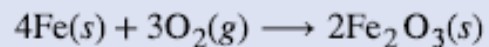
How does the spontaneity of this process depend upon temperature?

Solution

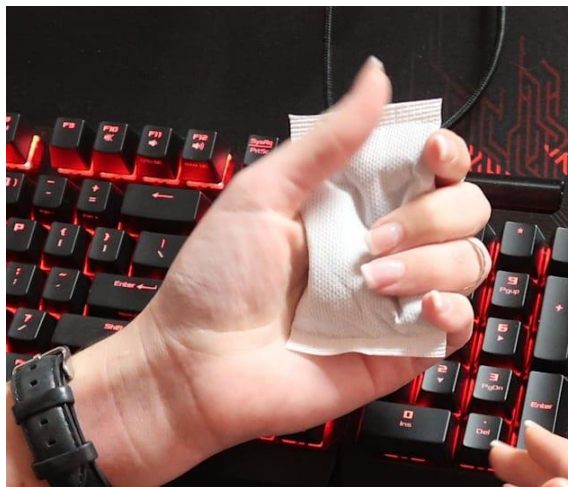
Combustion processes are exothermic ($\Delta H < 0$). This particular reaction involves an increase in entropy due to the accompanying increase in the amount of gaseous species (net gain of one mole of gas, $\Delta S > 0$). The reaction is therefore spontaneous ($\Delta G < 0$) at all temperatures.

Check Your Learning

Popular chemical hand warmers generate heat by the air-oxidation of iron:



How does the spontaneity of this process depend upon temperature?



Answer: ΔH and ΔS are negative; the reaction is spontaneous at low temperatures.

